PHYS3071

# Introduction to Astrophysics

Course Notes

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Compiled: *Tuesday 2nd June, 2026*

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# Introduction

## About These Notes

These notes accompany the PHYS3071 Astrophysics course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in basic physics, basic linear algebra, and basic calculus.

The material covers different astrophysical topics, including but not limited to the formation and evolution of stars, the type and spectrum of stars. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

*These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.*

If you find any errors or have suggestions for improvement, please feel free to contact me at `yhip@connect.ust.hk`. You can check the latest version of the notes on GitHub: https://github.com/AuTomatoc-Ti/phys3071-Astrophysics-notes/blob/main/main.pdf.

## Document Conventions

### Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

> **PHYSICS INTUITION**
> **Purpose:** Builds conceptual understanding through plain language explanations.
> This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.
> *Use this when:* You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

### Theory

**Purpose:** Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

*Use this when:* You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

### Question Examples

**Purpose:** Demonstrates practical application through worked problems.
Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

*Use this when:* Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

### Foundation

**Purpose:** Supplies background knowledge and detailed proof steps that underpin the main content.
This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

*Use this when:* A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

### Extra

**Purpose:** Provides supplementary material that extends *beyond* the course syllabus.
This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

*Use this when:* You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

**Suggested Reading Paths:**

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

## Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

**Important Note**

**Purpose:** Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.
An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

*Treat these as mandatory reading* regardless of which reading path you follow.

**Remark**

**Purpose:** Highlights a useful observation, connection, or naming convention.
A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

*Use this when:* Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

## Mathematical Conventions

**Key symbols.**

| Symbol | Meaning | Typical units |
|---|---|---|
| $M_\odot$ | Solar mass | kg |
| $R_\odot$ | Solar radius | m |
| $L_\odot$ | Solar luminosity | W |
| $T_{\text{eff}}$ | Effective temperature | K |
| $\mu$ | Mean molecular weight | — |
| $\rho$ | Mass density | $\text{kg}\,\text{m}^{-3}$ |

**Remark**

Unless otherwise stated, stellar quantities are quoted in solar-normalised form (for example $M/M_\odot$ and $L/L_\odot$), which makes scaling relations easier to compare across different stellar masses.

**differential notation.** We use $\dot{x} \equiv \frac{dx}{dt}$ for time derivatives of some variables $x(t)$.

CHAPTER 1

# Basic Concepts in Astrophysics

Astrophysics is the branch of astronomy that applies the principles of physics and chemistry to understand the nature of celestial objects and phenomena. It includes but is not limited to the study of stars, galaxies, extrasolar planets, cosmic microwave background radiation, and the interstellar medium. In this notes, we will only focus on the field of stellar astrophysics, which is the study of stars, their properties, structure, types, and evolution.

## 1.1 Stars and Stellar Measurements

**Physics Intuition**

In this notes, we define a star as a celestial object that is:

- (1) bounded by self-gravity under hydrostatic equilibrium, and
- (2) radiating energy generated by internal sources (e.g. nuclear fusion).

For most of the case, a star is spherical due to spherical symmetry of gravity, and mainly radiating energy by nuclear fusion in the core. However, we will also encounter some exceptions (e.g. energy generated by gravitational contraction in late stages of star evolution).

This definition shows that other celestial objects (e.g. planets, comets) are not stars as they don't fulfill all conditions. On the other hand, it allow some special conditions (e.g. brown dwarfs, white dwarfs) to be classified as stars, as they are still self-gravitating and radiating energy (though not by nuclear fusion). However, be careful some astrophysicists may not treat them as "real stars" as they don't have nuclear fusion, but we will still include them in our discussion as they are important for understanding the full picture of stellar evolution.

**Remark**

The stars are usually of spherical shape due to rotational symmetric force field of gravity, but can be oblate if they rotate rapidly.

### 1.1.1 Distance and Parallax

**Physics Intuition**

Parallax is the apparent shift in position of a nearby star against the background of distant stars when observed from different positions (e.g. Earth at different points in its orbit). By measuring this shift, we can determine the distance to the star using simple trigonometry.

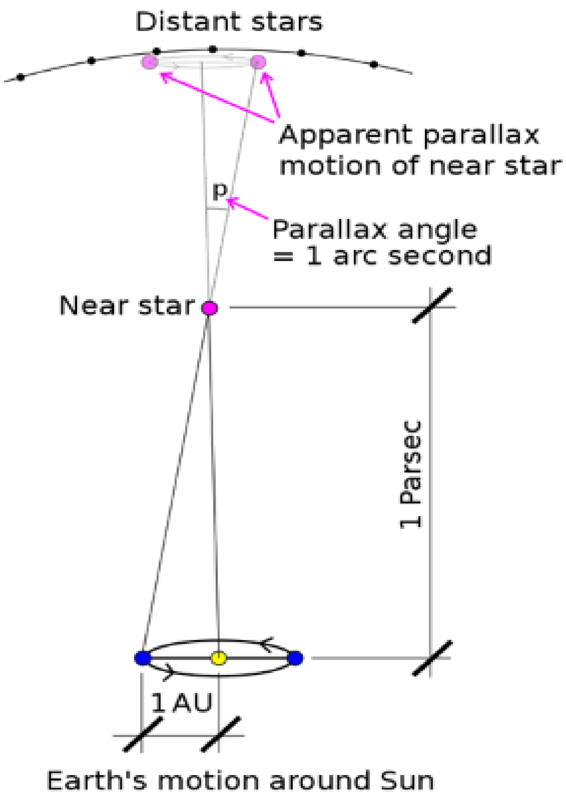


Figure 1.1: Stellar parallax illustration. The nearby star appears to shift position against the background of distant stars as Earth orbits the Sun. The angle $p$ is the shifted angle actually, instead of the parallex angle $\frac{1}{2}p$.

**Theory**

Let x be the baseline (e.g. diameter of Earth's orbit), d be the distance to the star, and $\theta$ be the parallax angle (half of the shifted angle $p$). Then we have:

- Parallax distance:
$$d_{\text{pc}} = \tan^{-1}\left(\frac{x/2}{d}\right) \approx \frac{1\text{AU}}{\theta_{\text{arcsec}}} \tag{1.1}$$
- we can also directly calculate the radius of the star if we know its angular radius $\theta$ and distance $d$:
$$R = d\tan\theta \approx d\theta \tag{1.2}$$

**Important Note**

Don't mixed up the parallax angle $\theta$ and the shifted angle $p$ in the parallax measurement. The total shifted angle $p$ shows the apparent shift of the star, but when we use the trigonometry to calculate the distance, we need to use the parallax angle $\theta = \frac{1}{2}p$ which is the angle between the line of sight to the star and the line connecting the Earth and the Sun.

**Question Example 1.1**

From the previous example, if the star has an angular diameter of 0.01 arcseconds, estimate its radius in units of $R_\odot$.

**Solution.**

$$\begin{aligned} R &= \frac{1}{2}\theta d = \frac{1}{2} \times 0.01'' \times 100\,\text{pc} \\ &= 0.5 \times 0.01 \times 100 \times 206265\,\text{AU} \\ &= 1031.325\,\text{AU} \\ &= 1031.325 \times 215 R_\odot \\ &\approx 221,000 R_\odot \end{aligned}$$

**Question Example 1.2**

Assume a telescope angular resolution of 0.01 arcseconds.
(a). what is the maximum distance at which we can resolve a star with a radius of $R_\odot$?
(b). If baseline $x$ is 2 AU, what is the maximum distance of a star we can resolve?

**Solution.** (a). We can use the formula for the radius of the star:

$$\begin{aligned} R &\geq d\theta \\ d &\leq \frac{R}{\theta} = \frac{R_\odot}{0.01''} \end{aligned}$$

As $R_\odot \approx 6.957 \times 10^8$ m and $1'' = 4.848 \times 10^{-6}$ radians, we have:

$$\begin{aligned} d &\leq \frac{6.957 \times 10^8\,\text{m}}{0.01 \times 4.848 \times 10^{-6}} \\ &\approx 1.43 \times 10^{16}\,\text{m} \\ &\approx 0.465\,\text{pc} \end{aligned}$$

(b). Using the parallax distance formula:

$$\begin{aligned} d &\leq \frac{x/2}{\tan\theta} \approx \frac{1\,\text{AU}}{0.01''} \\ &\approx 100\,\text{pc} \end{aligned}$$

**Remark**

As we can see from the above examples, the stellar parallax method is only effective for measuring distances to nearby stars (within a few hundred parsecs). For more distant stars, we need to use other methods such as spectroscopic parallax or standard candles.(we will discuss these methods in later chapters).

### 1.1.2 Brightness, Magnitude, and Luminosity

**Physics Intuition**

The brightness of a star is how bright it appears to us on Earth, in ancient times, it was measured by the naked eye and categorized into 6 magnitudes (1st magnitude being the brightest and 6th magnitude being the faintest).

In modern astronomy, we keep the magnitude system but use more precise definitions and measurements. we define the magnitude system as logarithmic, meaning that a difference of 5 magnitudes corresponds to a factor of 100 in brightness. And the logarithmic scale also allows a star to be brighter than 1st magnitude (e.g. Sirius has a magnitude of -1.46) or fainter than 6th magnitude (e.g. pluto has a magnitude of 14.0).

**Theory**

The **apparent magnitude** $m$ of a star is defined as:

$$m = -2.512 \log_{10} \left( \frac{F}{F_0} \right) \tag{1.3}$$

where $F$ is the observed flux from the star and $F_0$ is a reference flux (the flux of a star with magnitude 0). Some astrophysicists also use $B$ to represent the observed flux as "brightness". And we can write the formula more generally as:

$$m_1 - m_2 = -2.512 \log_{10} \left( \frac{F_1}{F_2} \right) \tag{1.4}$$

where $m_1$ and $m_2$ are the apparent magnitudes of two stars, and $F_1$ and $F_2$ are their corresponding fluxes.

**Physics Intuition**

One may ask that a dimmer star closer to us can have the same apparent magnitude as a brighter star farther away. To compare the intrinsic brightness of stars, we need to use the concept of **absolute magnitude**, which is defined as the apparent magnitude a star would have if it were located at a standard distance of 10 parsecs from Earth.

**Foundation**

The absolute magnitude $M$ of a star is defined as:

$$M_2 - M_1 = -2.512 \log_{10}\left(\frac{L_2}{L_1}\right)$$

and relationship between flux $F$, luminosity $L$ is:

$$F = \frac{L}{4\pi d^2}$$

hence we consider the 2 situations:

- case 1: star at 10 pc, we have $m_1 = M_1$ and $F_1 = \frac{L}{4\pi(10\,\text{pc})^2}$
- case 2: same star at distance $d$, we have $m$ and $F_2 = \frac{L}{4\pi d^2}$

Then we can derive the distance modulus formula:

$$\begin{aligned}
m_2 - m_1 &= -2.512 \log_{10}\left(\frac{F_2}{F_1}\right) \\
m_2 - M &= -2.512 \log_{10}\left(\frac{L/4\pi d^2}{L/4\pi(10\,\text{pc})^2}\right) \\
m_2 - M &= -2.512 \log_{10}\left(\frac{\frac{1}{d^2}}{\frac{1}{(10\,\text{pc})^2}}\right) \\
m_2 - M &= -2.512 \log_{10}\left(\frac{(10\,\text{pc})^2}{d^2}\right) \\
m_2 - M &= 5 \log_{10}\left(\frac{d}{10\,\text{pc}}\right)
\end{aligned}$$

**Theory**

We know that the relationship between the flux $F$ and luminosity $L$ of a star is given by:

$$F = \frac{L}{4\pi d^2} \tag{1.5}$$

where $d$ is the distance to the star.

And hence the **absolute magnitude** $M$ of a star is defined as:

$$M_2 - M_1 = -2.512 \log_{10} \left( \frac{L_2}{L_1} \right) \tag{1.6}$$

We can further relate the apparent magnitude $m$, absolute magnitude $M$, and distance $d$ using the distance modulus formula:

$$m - M = 5 \log_{10} \left( \frac{d}{10\text{pc}} \right) \tag{1.7}$$

**Question Example 1.3: Distance modulus quick use**

A star has $m = 9.0$ and $M = 4.0$. Estimate its distance.

**Solution.**

$$\begin{aligned}
m - M &= 5 \log_{10} \frac{d}{10\text{pc}} \\
9.0 - 4.0 &= 5 \log_{10} \frac{d}{10\text{pc}} \\
5.0 &= 5 \log_{10} \frac{d}{10\text{pc}} \\
\log_{10} \frac{d}{10\text{pc}} &= 1.0 \\
\frac{d}{10\text{pc}} &= 10^{1.0} = 10 \\
d &= 100\text{pc}
\end{aligned}$$

## 1.2 Thermal Radiation and Blackbody Spectrum

### 1.2.1 Planck function, Rayleigh-Jeans law, Wien's law

Classically, the Rayleigh-Jeans law describes the spectral energy distribution of thermal radiation at low frequencies (long wavelengths):

$$B_\nu(T) \approx \frac{2k_B T \nu^2}{c^2} \quad \text{for } h\nu \ll k_B T \tag{1.8}$$

However, this law fails at high frequencies (short wavelengths), leading to the **ultraviolet catastrophe**. The Planck function resolves this issue by introducing quantization of energy levels:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \tag{1.9}$$

**Physics Intuition**

A star's spectrum can be approximated as a blackbody, which is an idealized object that absorbs all incident radiation and re-emits it with a characteristic spectrum determined solely by its temperature. The Planck function describes the spectral energy distribution of a blackbody, while the Stefan-Boltzmann law relates the total energy emitted to the temperature. Wien's law gives the wavelength at which the emission peaks, allowing us to estimate the star's surface temperature from its observed spectrum.

**Theory**

- Planck function:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \tag{1.10}$$

- Stefan–Boltzmann law:

$$F = \sigma T^4, \quad \sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}} \tag{1.11}$$

- Luminosity:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4 \tag{1.12}$$

- Wien's law:

$$\lambda_{\max} T = 2.9 \times 10^{-3}\,\mathrm{m\,K} \tag{1.13}$$

**Question Example 1.4**

A star has a peak wavelength of 500 nm. Estimate its surface temperature.

**Solution.** From Wien's law,

$$\begin{aligned}\lambda_{\max} T &= 2.9 \times 10^{-3}\,\mathrm{m\,K} \\ T &= \frac{2.9 \times 10^{-3}\,\mathrm{m\,K}}{\lambda_{\max}} \\ &= \frac{2.9 \times 10^{-3}\,\mathrm{m\,K}}{500 \times 10^{-9}\,\mathrm{m}} \\ &= 5800\,\mathrm{K}.\end{aligned}$$

**Question Example 1.5**

A star has a surface temperature of 6000 K and a radius of $2R_\odot$. Estimate its luminosity in units of $L_\odot$.

**Solution.** From the Stefan-Boltzmann law,

$$\begin{aligned}L &= 4\pi R^2 \sigma T_{\text{eff}}^4 \\ &= 4\pi (2R_\odot)^2 \sigma (6000\,\mathrm{K})^4 \\ &= 16\pi R_\odot^2 \sigma (6000\,\mathrm{K})^4 \\ &= 16 L_\odot \left(\frac{6000}{5778}\right)^4 \\ &\approx 16 L_\odot.\end{aligned}$$

**Question Example 1.6**

Calculate the specific intensity of a blackbody at $T = 5000\,\mathrm{K}$ at a frequency of $\nu = 5 \times 10^{14}\,\mathrm{Hz}$.

**Solution.** Using the Planck function,

$$\begin{aligned}B_\nu(T) &= \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \\ &= \frac{2 \times 6.626 \times 10^{-34} \times (5 \times 10^{14})^3}{(3 \times 10^8)^2} \frac{1}{e^{(6.626 \times 10^{-34} \times 5 \times 10^{14})/(1.38 \times 10^{-23} \times 5000)} - 1} \\ &\approx 1.2 \times 10^{-5}\,\mathrm{W/m^2 Hz\,sr}.\end{aligned}$$

### 1.2.2 Intensity, Flux, and Luminosity

**Physics Intuition**

Observed starlight is measured in layers: direction-resolved intensity, surface flux, and finally total emitted power. Keeping these definitions separate avoids unit mistakes in later derivations.

**THEORY**

- Specific intensity $I_\nu$:
$$dE = I_\nu \cos\theta \, dA \, dt \, d\nu \, d\Omega \tag{1.14}$$
- Monochromatic flux density:
$$F_\nu = \int I_\nu \cos\theta \, d\Omega \tag{1.15}$$
- Bolometric flux:
$$F = \int_0^\infty F_\nu \, d\nu \tag{1.16}$$
- Luminosity:
$$L = \int F \, dA \tag{1.17}$$

**IMPORTANT NOTE**

$I_\nu$ is invariant along a ray in vacuum (no absorption/emission), while $F_\nu$ depends on source geometry and distance. Do not interchange them in magnitude or distance calculations.

### 1.2.3 Radiative Transfer

**PHYSICS INTUITION**

Radiative transfer describes how light propagates through a medium, accounting for absorption and emission processes.

**THEORY**

The radiative transfer equation is given by:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu \tag{1.18}$$

where $\alpha_\nu$ is the absorption coefficient and $j_\nu$ is the emission coefficient. The solution to this equation can be expressed in terms of the optical depth $\tau_\nu$:

$$I_\nu(s) = I_\nu(0)e^{-\tau_\nu} + \int_0^s j_\nu(s')e^{-(\tau_\nu(s)-\tau_\nu(s'))}ds' \tag{1.19}$$

where $\tau_\nu(s) = \int_0^s \alpha_\nu(s')ds'$ is the optical depth along the path.

**Question Example 1.7: Calculating Brightness**

Suppose a star is at 20 pc. How bright does it appear if its luminosity is $100L_\odot$?
**Solution.**

$$\begin{aligned} F &= \frac{L}{4\pi d^2} = \frac{100L_\odot}{4\pi(20\,\mathrm{pc})^2} \\ &= \frac{100 \times 3.828 \times 10^{26}\,\mathrm{W}}{4\pi(20 \times 3.086 \times 10^{16}\,\mathrm{m})^2} \\ &\approx 3.1 \times 10^{-9}\,\mathrm{W/m^2}. \end{aligned}$$

### 1.2.4 Surface Temperature

Recall that the surface temperature $T_{\text{eff}}$ of a star can be estimated from its luminosity and radius using the Stefan-Boltzmann law:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4 \tag{1.20}$$

where $\sigma$ is the Stefan-Boltzmann constant. Rearranging this gives:

$$T_{\text{eff}} = \left(\frac{L}{4\pi R^2 \sigma}\right)^{1/4} \tag{1.21}$$

**Question Example 1.8: Estimating Surface Temperature**

A star has a luminosity of $300L_\odot$ and a radius of $25R_\odot$. Estimate its surface temperature.
**Solution.** First, we need to convert the luminosity and radius into SI units:

$$\begin{aligned} L &= 300L_\odot = 300 \times 3.828 \times 10^{26}\,\mathrm{W} = 1.1484 \times 10^{29}\,\mathrm{W}, \\ R &= 25R_\odot = 25 \times 6.957 \times 10^{8}\,\mathrm{m} = 1.73925 \times 10^{10}\,\mathrm{m}. \end{aligned}$$

Now we can plug these values into the formula for $T_{\text{eff}}$:

$$\begin{aligned} T_{\text{eff}} &= \left(\frac{1.1484 \times 10^{29}\,\mathrm{W}}{4\pi(1.73925 \times 10^{10}\,\mathrm{m})^2 \times 5.67 \times 10^{-8}\,\mathrm{W/m^2K^4}}\right)^{1/4} \\ &\approx 4500\,\mathrm{K}. \end{aligned}$$

CHAPTER 2

# Star Formation

## 2.1 Sources of Star Formation

### 2.1.1 Molecular Clouds

PHYSICS INTUITION

Molecular clouds are dense regions of gas and dust in the interstellar medium where star formation occurs. They are primarily composed of molecular hydrogen ($H_2$) and can have masses ranging from a few solar masses to millions of solar masses. The high density and low temperature of molecular clouds allow them to collapse under their own gravity, leading to the formation of stars.

## 2.2 Analysis of Formation Criteria

### 2.2.1 Hydrostatic Equilibrium

FOUNDATION

Assume spherical symmetry and radial dependence only. Neglect rotation, magnetic fields, and turbulence in the first model to isolate gravity-versus-pressure physics.

THEORY

$$dm(r,t) = 4\pi r^2 \rho \, dr, \tag{2.1}$$

$$\frac{\partial m}{\partial t} + v\frac{\partial m}{\partial r} = 0. \tag{2.2}$$

**Question Example 2.1**

Derive the radial acceleration $\ddot{r}$ of a fluid element at radius $r$ in a star, given the mass enclosed within $r$ is $m(r)$ and the density is $\rho(r)$.

**Solution.** The gravitational force per unit mass at radius $r$ is given by:

$$g = \frac{Gm(r)}{r^2}. \tag{2.3}$$

The pressure gradient force per unit mass is given by:

$$f_P = -\frac{1}{\rho}\frac{dP}{dr}. \tag{2.4}$$

The total radial acceleration $\ddot{r}$ is the sum of these two forces:

$$\ddot{r} = f_P + g = -\frac{1}{\rho}\frac{dP}{dr} + \frac{Gm(r)}{r^2}. \tag{2.5}$$

**Question Example 2.2**

Consider a star of mass $M$ and radius $R$ in hydrostatic equilibrium. If the star is suddenly compressed to half its radius, what is the radial acceleration $\ddot{r}$ at the surface?

**Solution.** The gravitational force per unit mass at the surface is given by:

$$g = \frac{GM}{R^2}. \tag{2.6}$$

If the star is compressed to half its radius, the new radius is $R' = R/2$. The new gravitational force per unit mass at the surface is:

$$g' = \frac{GM}{(R/2)^2} = \frac{GM}{R^2/4} = 4\frac{GM}{R^2} = 4g. \tag{2.7}$$

The radial acceleration $\ddot{r}$ at the surface is then:

$$\ddot{r} = g' - g = 3g = 3\frac{GM}{R^2}. \tag{2.8}$$

**Theory**

Hydrostatic equilibrium At hydrostatic equilibrium, the inward gravitational force is balanced by the outward pressure gradient and $\ddot{r} = 0$:

$$\frac{dP}{dr} = -\frac{Gm(r)\rho}{r^2} \tag{2.9}$$

**Question Example 2.3**

Consider a sphere of mass $M$ and radius $R$ with a uniform density $\rho$.

(a) Write the pressure gradient $-\dfrac{dP}{dr}$ at distance $r$ from the center at hydrostatic equilibrium.

(b) Calculate the pressure $P(r)$ at distance $r$ from the center.

**Solution.**

(a) The mass enclosed within radius $r$ is given by:

$$m(r) = \frac{4}{3}\pi r^3 \rho. \tag{2.10}$$

The hydrostatic equilibrium equation is:

$$-\frac{dP}{dr} = \frac{Gm(r)\rho}{r^2}. \tag{2.11}$$

Substituting $m(r)$, we get:

$$-\frac{dP}{dr} = \frac{G\left(\frac{4}{3}\pi r^3 \rho\right)\rho}{r^2} = \frac{4\pi G}{3} r \rho^2. \tag{2.12}$$

(b) To find $P(r)$, we integrate the pressure gradient from the surface ($r = R$, where $P(R) = 0$) to a point inside the sphere:

$$\begin{aligned}
P(r) &= -\int_r^R -\frac{dP}{dr'}dr' = \int_r^R \frac{4\pi G}{3} r' \rho^2 dr' \\
&= \frac{4\pi G}{3}\rho^2 \int_r^R r' dr' = \frac{4\pi G}{3}\rho^2 \left[\frac{r'^2}{2}\right]_r^R \\
&= \frac{4\pi G}{3}\rho^2 \left(\frac{R^2 - r^2}{2}\right) = 2\pi G \rho^2 (R^2 - r^2).
\end{aligned}$$

## 2.3 Virial Theorem and Jeans Criterion

### 2.3.1 Energetics of Self-Gravitating Gas

**Theory**

$$E_{\mathrm{GR}} = -\int_0^M \frac{Gm}{r}\,dm = -f_{\mathrm{GR}}\frac{GM^2}{R}, \tag{2.13}$$

$$\langle P \rangle = -\frac{1}{3}\frac{E_{\mathrm{GR}}}{V}, \tag{2.14}$$

$$P = \frac{1}{3}nm\langle v^2 \rangle = \frac{2}{3}n\langle E_k \rangle. \tag{2.15}$$

For a star with a uniform density profile (i.e., $\rho$ = constant), $f_{\mathrm{GR}} \approx 3/5$.

### 2.3.2 Jeans Criterion

**Physics Intuition**

The Jeans criterion determines whether a cloud of gas will collapse under its own gravity to form a star. It compares the gravitational energy of the cloud to its thermal energy. If the gravitational energy exceeds twice the thermal energy, the cloud is unstable and will collapse.

### 2.3.3 Jeans Instability

**Theory**

$$|E_{\mathrm{GR}}| > 2E_{\mathrm{th}} \quad \text{(collapse)}, \tag{2.16}$$

$$M_J = \left(\frac{5k_B T}{G\mu m_H}\right)^{3/2}\left(\frac{3}{4\pi\rho}\right)^{1/2}, \tag{2.17}$$

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G\rho}}, \tag{2.18}$$

$$M_J \propto T^{3/2}\rho^{-1/2}. \tag{2.19}$$

**Important Note**

Lower temperature and higher density both reduce $M_J$. This is why cooling strongly promotes fragmentation and cluster formation.

**Question Example 2.4**

Calculate the Jeans mass for a molecular cloud with a temperature of 10 K and a density of $10^{-20}\,\mathrm{kg/m^3}$. Assume the mean molecular weight $\mu$ is 2.3 (typical for molecular clouds).

**Solution.** First, we need to calculate the sound speed $c_s$ using the formula:

$$c_s = \sqrt{\frac{k_B T}{\mu m_H}}, \tag{2.20}$$

where $k_B$ is the Boltzmann constant ($1.38 \times 10^{-23}\,\mathrm{J/K}$), $T$ is the temperature, $\mu$ is the mean molecular weight, and $m_H$ is the mass of a hydrogen atom ($1.67 \times 10^{-27}\,\mathrm{kg}$). Substituting the values, we get:

$$c_s = \sqrt{\frac{1.38 \times 10^{-23} \times 10}{2.3 \times 1.67 \times 10^{-27}}} \approx 0.19\,\mathrm{km/s}. \tag{2.21}$$

Next, we can calculate the Jeans mass $M_J$ using the formula:

$$M_J = \left(\frac{5k_B T}{G\mu m_H}\right)^{3/2} \left(\frac{3}{4\pi\rho}\right)^{1/2}, \tag{2.22}$$

where $G$ is the gravitational constant ($6.67 \times 10^{-11}\,\mathrm{N\,m^2/kg^2}$) and $\rho$ is the density. Substituting the values, we get:

$$\begin{aligned} M_J &= \left(\frac{5 \times 1.38 \times 10^{-23} \times 10}{6.67 \times 10^{-11} \times 2.3 \times 1.67 \times 10^{-27}}\right)^{3/2} \left(\frac{3}{4\pi \times 10^{-20}}\right)^{1/2} && (2.23) \\ &\approx 1.0 \times 10^{30}\,\mathrm{kg} \approx 0.5\,M_\odot. && (2.24) \end{aligned}$$

## 2.4 End of Formation: Protostars and Pre-Main Sequence Stars

CHAPTER 3

# Thermodynamic Properties of Stars

## 3.1 Statistical Distributions

### 3.1.1 Energy-Momentum Relation

To calculate the value of various thermodynamic quantities, we need first clarify some basic concepts and notations first for convenience.

In below contents, we will use the $E(p)$ notation to represent the energy of a particle with momentum $p$ and mass $m$.

**FOUNDATION**

Recall special relativity, the energy of a particle with mass $m$ and three-momentum $p$ is given by

$$E(\vec{p}) = \sqrt{|\vec{p}|^2c^2 + m^2c^4}.$$

Taken $c = 1$ and $p = |\vec{p}|$, we have $E(p) = \sqrt{p^2 + m^2}$.

In a Non-relativistic limit ($p \ll m$):

$$E(p) = m\sqrt{1 + \frac{p^2}{m^2}} \approx m + \frac{p^2}{2m}.$$

In an Ultra-relativistic limit ($p \gg m$):

$$E(p) \approx p.$$

### 3.1.2 Chemical Potential

**PHYSICS INTUITION**

Imagine we have a box of gas, and we want to add a particle into the box. The chemical potential $\mu$ is the energy cost of adding that particle to the system, which can be thought of as the "price" we need to pay to add one more particle to the system.

However, when we add a particle to the system, we also increase the entropy of the system, which can be thought of as the "benefit" we get from adding that particle, since it allows for more possible microstates and thus increases the disorder of the system.

Furthermore, the temperature $T$ of the system determines how much we value that increase in entropy, since at higher temperatures, we need to pay more energy to achieve the same increase in entropy.

Therefore, the chemical potential can be thought of as a balance between the energy cost of adding a particle and the entropy gain from adding that particle, which is given by the formula:

$$\mu = \text{energy cost} - T \cdot \text{entropy gain}$$

**FOUNDATION**

$\mu$ is the chemical potential, which measures the change in internal energy when a particle is added to the system at constant entropy and volume. It is defined as:

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} = \left(\frac{\partial F}{\partial N}\right)_{T,V},$$

Or in other way:

$$\delta\mu = \delta U - T\delta S,$$

where $U$ is the internal energy, $F$ is the Helmholtz free energy, $N$ is the number of particles, $S$ is the entropy, and $T$ is the temperature.

For ideal gases, $\mu \to 0$, and we will assume $\mu = 0$ in below contents unless otherwise specified.

**REMARK**

The chemical potential $\mu$ is typically zero for photons and other massless bosons, while it can be non-zero for massive particles depending on the context (e.g., in a gas of electrons).

while for particles and their own anti-particles, $\mu_+ = -\mu_-$.

### 3.1.3 Thermal Equilibrium Distributions

**FOUNDATION**

For a state with energy $E$, and particle number $N$, the probability of the state is given by the Boltzmann factor:

$$P \propto e^{-(E-\mu N)/k_B T}$$

While for a single-particle state with energy $E(p)$, each momentum state can have occupation number $n$:

- Fermions: $n = 0$ or $1$ ,
- Bosons: $n = 0, 1, 2, \ldots$

Then the average occupation number for a single-particle state with energy $E(p)$ is given by:

$$\langle n \rangle = \frac{\sum_n n e^{-(E(p)-\mu n)/k_B T}}{\sum_n e^{-(E(p)-\mu n)/k_B T}}$$

For fermions, we have:

$$\langle n \rangle = \frac{0 \cdot e^{-E(p)/k_B T} + 1 \cdot e^{-(E(p)-\mu)/k_B T}}{e^{-E(p)/k_B T} + e^{-(E(p)-\mu)/k_B T}} = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

For bosons, we have:

$$\langle n \rangle = \frac{\sum_{n=0}^{\infty} n e^{-(E(p)-\mu n)/k_B T}}{\sum_{n=0}^{\infty} e^{-(E(p)-\mu n)/k_B T}} = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

By letting $f(p) = \langle n \rangle$, we can calculate the number density by density of states $\times$ occupation number, which is given by:

$$\frac{dN}{d^3 p} = \overbrace{\frac{g}{(2\pi)^3}}^{\text{density of states}} \overbrace{f(p)}^{\text{occupation number}} = \frac{g}{(2\pi)^3} \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1}$$

The term $\frac{g}{2\pi^2}$ come from the Heisenberg uncertainty principle, which states that density states, i.e. the number of quantum states in a given volume of phase space as $(2\pi\hbar)^3$ under natural unit, and the factor $g$ accounts for the spin degeneracy of the particle species.

**THEORY**

In thermal equilibrium, the distribution of particle energies follows specific statistical distributions:

- **Fermi-Dirac distribution** for fermions:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

- **Bose-Einstein distribution** for bosons:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

- **Maxwell-Boltzmann distribution** for classical particles:

$$f(p) = e^{-(E(p)-\mu)/k_B T}$$

In general, the distribution can be written as:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1},$$

where the $+$ sign is for fermions and the $-$ sign is for bosons.

**THEORY**

In non-relativistic limit($E \gg T$), the Fermi-Dirac and Bose-Einstein distributions reduce to the Maxwell-Boltzmann distribution:

$$e^{(E(p)-\mu)/k_B T} \gg 1 \implies f(p) \approx e^{-(E(p)-\mu)/k_B T}$$

## 3.2 Number Density, Energy Density, and Pressure

### 3.2.1 Spin Degeneracy Factor

**Physics Intuition**

Imagine in a hotel, there are multiple rooms with different price(Energy level), the spin is like the no. of beds in the room, which can accommodate multiple guests(particles) with the same energy level. When 2 particles enter the same room(share the same energy level), we call it a degenerate state.

The degeneracy factor $g$ states how many particles can stay in the same energy level, which is determined by the intrinsic spin quantum number $s$ of the particle. On the other hand, it can be treat as the degree of freedom of the particle, which is the number of independent ways the particle can exist in a given energy state.

**Foundation**

In quantum mechanics, the magnetic quantum number $m_s$ can take on values from $-s$ to $+s$ in integer steps($-s, -s+1, \ldots, s-1, s$). And we can get $g$ by counting the number of possible $m_s$ values, which is given by:

$$g = 2s + 1$$

However, for massless particles, such as photons, the spin is always aligned with the direction of motion, which means they only have 2 possible spin states (helicity states), thus $g = 2$ for photons.

**Remark**

The $g$ here only counts the intrinsic spin degeneracy, but in general, we also need to consider other types of degeneracy, such as color charge for quarks, which can further increase the effective degeneracy factor for those particles: For example, if we treat the 8 type of gulons as the same one and calculate the $g$ for gluons, we will get $g_{\text{total}} = 16$.

$$g_{\text{effective}} = g_{\text{spin}} \times g_{\text{color}}$$

In the below contents, when we see $g_i$, it only represents the intrinsic spin degeneracy, and we will need to multiply it with other degeneracy factors if there are any.

EXTRA

The g values for some common particles are:

| Particle | symbol | $g$ |
|---|---|---|
| Photon | $g_\gamma$ | 2 |
| Electron/Positron | $g_e^-, g_e^+$ | 2 |
| Z boson | $g_Z$ | 3 |
| Higgs boson | $g_H$ | 1 |

### 3.2.2 Derivation of the 3 equations

THEORY

The number density $n$, energy density $\rho$, and pressure $P$ of some particle i are given by:

$$n_i = \frac{g_i}{(2\pi)^3} \int f_i(p)\, d^3p,$$

$$\rho_i = \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p,$$

$$P_i = \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p$$

By substituting the Fermi-Dirac, Bose-Einstein, or Maxwell-Boltzmann distribution functions into the above integrals, we can derive explicit expressions for $n_i$, $\rho_i$, and $P_i$ for different particle species under various conditions (e.g., relativistic vs. non-relativistic limits).

FOUNDATION

**(1) Relativistic Fermions ($E \approx p$):**

- Number density:

$$\begin{aligned} n_i &= \frac{g_i}{(2\pi)^3} \int \frac{d^3p}{e^{(E(p)-\mu)/k_BT}+1} \\ &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{4\pi p^2 dp}{e^{(p-\mu)/k_BT}+1} && \text{, using spherical coordinates} \\ &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \int_0^\infty \frac{u^2 du}{e^u+1} && , u = \frac{p-\mu}{k_BT} \\ &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \cdot \frac{3}{2}\zeta(3) \\ &= \frac{3}{4}\frac{\zeta(3)}{\pi^2} g_i T^3 \end{aligned}$$

- Energy density:

$$\begin{aligned} \rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p \\ &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT}+1} && \text{, E = p for relativistic particles} \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u+1} && , u = \frac{p-\mu}{k_BT} \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot \frac{7}{8} \cdot 6\zeta(4) \\ &= \frac{7}{8}\frac{\pi^2}{30} g_i T^4 \end{aligned}$$

- Pressure:
  using:

$$\frac{p^2}{3E(p)} = \frac{1}{3}\frac{p^2}{p} = \frac{1}{3}p,$$

  we get:

$$\begin{aligned} P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p \\ &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{\frac{1}{3}p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT}+1} \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u+1} && , u = \frac{p-\mu}{k_BT} \\ &= \frac{7}{8}\frac{\pi^2}{30} g_i T^4 \cdot \frac{1}{3} \\ &= \frac{1}{3}\rho_i \end{aligned}$$

**(2) Relativistic Bosons ($E \approx p$):**

- Number density:

$$\begin{aligned} n_i &= \frac{g_i}{(2\pi)^3}\int \frac{d^3p}{e^{(E(p)-\mu)/k_BT}-1} \\ &= \frac{g_i}{(2\pi)^3}\int_0^\infty \frac{4\pi p^2 dp}{e^{(p-\mu)/k_BT}-1} \\ &= \frac{g_i}{2(\pi)^2}k_B^3T^3\int_0^\infty \frac{u^2du}{e^u-1} & ,u=\frac{p-\mu}{k_BT} \\ &= \frac{g_i}{2(\pi)^2}k_B^3T^3\cdot 2\zeta(3) \\ &= \frac{\zeta(3)}{\pi^2}g_iT^3 \end{aligned}$$

- Energy density:
  using $E(p) \approx p$, we have:

$$\begin{aligned} \rho_i &= \frac{g_i}{(2\pi)^3}\int E(p)f_i(p)\,d^3p \\ &= \frac{g_i}{(2\pi)^3}\int_0^\infty \frac{p\cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT}-1} \\ &= \frac{g_i}{2(\pi)^2}k_B^4T^4\int_0^\infty \frac{u^3du}{e^u-1} & ,u=\frac{p-\mu}{k_BT} \\ &= \frac{g_i}{2(\pi)^2}k_B^4T^4\cdot 6\zeta(4) \\ &= \frac{g_i}{2(\pi)^2}k_B^4T^4\cdot\frac{\pi^4}{15} \\ &= \frac{\pi^2}{30}g_iT^4 \end{aligned}$$

- Pressure:

$$\begin{aligned} P_i &= \frac{g_i}{(2\pi)^3}\int \frac{p^2}{3E(p)}f_i(p)\,d^3p \\ &= \frac{g_i}{(2\pi)^3}\int_0^\infty \frac{\frac{1}{3}p\cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT}-1} \\ &= \frac{g_i}{2(\pi)^2}k_B^4T^4\int_0^\infty \frac{u^3du}{e^u-1} & ,u=\frac{p-\mu}{k_BT} \\ &= \frac{\pi^2}{30}g_iT^4\cdot\frac{1}{3} \\ &= \frac{1}{3}\rho_i \end{aligned}$$

**(3) Non-relativistic Particles ($E \approx m + p^2/2m$):**

- Number density:

$$\begin{aligned} n_i &= \frac{g_i}{(2\pi)^3} \int e^{-(E(p)-\mu)/k_BT} d^3p \\ &= \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_BT} \int e^{-p^2/2mk_BT} d^3p \\ &= \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_BT} \int 4\pi p^2 e^{-p^2/2mk_BT} dp \\ &= \frac{g_i}{2(\pi)^2} e^{-(m-\mu)/k_BT} \cdot \frac{\sqrt{\pi}}{4} (2mk_BT)^{3/2} \qquad \text{, using Gaussian integral} \\ &= g_i \left(\frac{mk_BT}{2\pi}\right)^{3/2} e^{-(m-\mu)/k_BT} \end{aligned}$$

- Energy density: using $E(p) \approx m + \frac{p^2}{2m}$, we have:

$$\begin{aligned} \rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p \\ &\approx \frac{g_i}{(2\pi)^3} \int \left(m + \frac{p^2}{2m}\right) e^{-(E(p)-\mu)/k_BT} d^3p \\ &= \frac{g_i m}{(2\pi)^3} e^{-(m-\mu)/k_BT} \int e^{-p^2/2mk_BT} d^3p \\ &\quad + \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_BT} \int \frac{p^2}{2m} e^{-p^2/2mk_BT} d^3p \\ &= n_i m + \frac{3}{2} n_i k_B T \qquad \text{, using Gaussian integral} \\ &= n_i m \qquad \text{, since } m \gg T \text{ for non-relativistic particles} \end{aligned}$$

- Pressure: using $\frac{p^2}{3E(p)} \approx \frac{p^2}{3m}$, we have:

$$\begin{aligned} P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p \\ &\approx \frac{g_i}{(2\pi)^3} \cdot \frac{2}{3} \int \frac{p^2}{2m} e^{-(E(p)-\mu)/k_BT} d^3p \\ &= n_i T \qquad \text{, by using the result from energy density calculation} \end{aligned}$$

**THEORY**

Number density, energy density, and pressure for particles with different thermal equilibrium distributions:

| | Relativistic Fermions | Relativistic Bosons | Non-relativistic Particles (Fermions/ Bosons) |
|---|---|---|---|
| Number Density $n_i$ | $\frac{3}{4}\frac{\zeta(3)}{\pi^2}g_i T^3$ | $\frac{\zeta(3)}{\pi^2}g_i T^3$ | $g_i\left(\frac{m_i T}{2\pi}\right)^{3/2}e^{-m_i/T}$ |
| Energy Density $\rho_i$ | $\frac{7}{8}\frac{\pi^2}{30}g_i T^4$ | $\frac{\pi^2}{30}g_i T^4$ | $n_i m_i$ |
| Pressure $P_i$ | $\frac{1}{3}\rho_i$ | $\frac{1}{3}\rho_i$ | $n_i T$ |

where $\zeta(3) \approx 1.202$ is the Riemann zeta function evaluated at 3.

We can see once the particles no longer relativistic, the number density and energy density will be exponentially suppressed.
It is because in the thermal plasma, the energy of particle are $E \sim T$, and when $T \ll 2m$, the probability of pair production which require energy $E \geq 2m$ will be very low, thus the number density and energy density will be suppressed.

**PHYSICS INTUITION**

It also explains why very few positrons exist in the current universe!
In the early universe, $T \gg m_e \approx 0.511\,\text{MeV}$, $e^+e^-$ pairs were abundant.
As the universe cooled below $T \sim m_e$, the number density of positrons (and electrons) dropped exponentially, leaving only a tiny fraction that survived annihilation, which is why we observe so few positrons today.

**PHYSICS INTUITION**

We can also observed the ferminos have lower number density and energy density than bosons.

Actually it also match our expectation in previous section, we know that bosons can occupied the same quantum state but fermions cannot (Pauli exclusion principle), thus we can expect that fermions have lower number density and energy density than bosons as lower occupation number.

**PHYSICS INTUITION**

For non-relativistic particles, we can see the energy density is dominated by the rest mass energy $n_i m_i$, while the pressure is given by $n_i T$, which is just the ideal gas law $P = nk_B T$ with $k_B = 1$ under natural unit.

**THEORY**

We can see the relationship between p and $\rho$ for different relativistic limits:

$$P = \begin{cases} \frac{1}{3}\rho & ,T \gg m \\ nT \ll \rho & ,T \ll m \end{cases}$$

In fact, it also consistent with the assumption of state parameter of radiation and matter, which are $w = 1/3$ and $w = 0$ when in relativistic and non-relativistic limits respectively.

**THEORY**

The average particle energy is given by:

$$\langle E \rangle = \frac{\rho_i}{n_i} = \begin{cases} \frac{7\pi^4}{180\zeta(3)}T \approx 3.151T & ,T \gg m \text{ for relativistic fermions} \\ \frac{\pi^4}{30\zeta(3)}T \approx 2.701T & ,T \gg m \text{ for relativistic bosons} \\ m + \frac{3}{2}T & ,T \ll m \text{ for non-relativistic particles} \end{cases}$$

Thats mean we can have a simplification of $\langle E \rangle \sim T$ for relativistic particles, as they have same order of magnitude, which is low enough error in astronomy :)

## 3.3 Statistical Distributions

**THEORY**

$$\text{Maxwell-Boltzmann}: \quad f_{\text{MB}}(E) = \frac{2\pi}{(k_B T)^{3/2}}\sqrt{E}e^{-E/k_B T}, \tag{3.1}$$

$$\text{Fermi-Dirac}: \quad f_{\text{FD}}(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}, \tag{3.2}$$

$$\text{Bose-Einstein}: \quad f_{\text{BE}}(E) = \frac{1}{e^{(E-\mu)/k_B T} - 1}. \tag{3.3}$$

**Question Example 3.1**

Calculate the most probable energy $E_{\text{mp}}$ for a Maxwell-Boltzmann distribution.

**Solution.** The most probable energy is found by setting the derivative of the distribution function to zero:

$$\frac{d}{dE}f_{\text{MB}}(E) = 0. \tag{3.4}$$

Solving this equation gives:

$$E_{\mathrm{mp}} = \frac{k_B T}{2}. \tag{3.5}$$

## 3.4 Equation of State (EOS) of Stellar Matter

**Physics Intuition**

The equation of state (EOS) relates the pressure, density, and temperature of a substance. For stellar matter, the EOS is crucial for understanding the internal structure and evolution of stars.

**Theory**

$$P = P(\rho, T, \{X_i\}), \tag{3.6}$$

$$\text{Ideal gas}: \quad P = \frac{\rho k_B T}{\mu m_H}, \tag{3.7}$$

$$\text{Radiation pressure}: \quad P_{\mathrm{rad}} = \frac{1}{3} a T^4, \quad a = \frac{4\sigma}{c}, \tag{3.8}$$

$$\text{Degenerate gas}: \quad P_{\mathrm{deg}} \propto \rho^{5/3} \quad \text{(non-relativistic)}, \tag{3.9}$$

$$P_{\mathrm{deg}} \propto \rho^{4/3} \quad \text{(relativistic)}. \tag{3.10}$$

## 3.5 Radiation

**Theory**

$$u_{\mathrm{rad}} = a T^4, \tag{3.11}$$

$$F_{\mathrm{rad}} = \frac{c}{4} u_{\mathrm{rad}} = \sigma T^4, \tag{3.12}$$

$$L_{\mathrm{rad}} = 4\pi R^2 F_{\mathrm{rad}} = 4\pi R^2 \sigma T^4. \tag{3.13}$$

CHAPTER 4

# Stellar Structure

Three main mechanisms of heat transfer in stars:

1. Conduction
2. Convection
3. Radiation (by photons and neutrinos from core)

## 4.1 Conduction

## 4.2 Convection

### 4.2.1 Convection Mechanism

## 4.3 Radiation

### 4.3.1 Scattering

### 4.3.2 Mean Free Path

**Theory**

The mean free path $\lambda$ of a photon in a medium is given by:

$$\lambda = \frac{1}{n\sigma}, \tag{4.1}$$

where $n$ is the number density of scatterers and $\sigma$ is the scattering cross-section.

**Question Example 4.1: Mean Free Path in the Sun's Interior**

alculate the mean free path of a photon in the Sun's interior, where the number density of electrons is approximately $n_e = 10^{26}\ \mathrm{m}^{-3}$ and the Thomson scattering cross-section is $\sigma_T = 6.65 \times 10^{-29}\ \mathrm{m}^2$.

**Solution:** Using the formula for mean free path:

$$\lambda = \frac{1}{n_e \sigma_T} = \frac{1}{(10^{26}\,\mathrm{m}^{-3})(6.65 \times 10^{-29}\,\mathrm{m}^2)} = 1.5 \times 10^3\,\mathrm{m}. \tag{4.2}$$

Thus, the mean free path of a photon in the Sun's interior is approximately 1.5 kilometers.

CHAPTER 5

# Nuclear Synthesis in Stars

Recall that there are three primary energy sources in stars: gravitational contraction, radioactive decay and nuclear fusion.

In this chapter, we will focus on the process of thermonuclear fusion, which is the viable long-term stellar energy source in most stars.

## 5.1 Requisites for Thermonuclear Fusion

### 5.1.1 Nuclear Size and Binding Energy

**THEORY**

$$R = R_0 A^{1/3}, \qquad R_0 \approx 1.2 \times 10^{-15}\,\mathrm{m}, \tag{5.1}$$

$$BE(Z,N) = \left[Z m_p + N m_n - M(Z,N)\right] c^2. \tag{5.2}$$

Binding energy per nucleon increases for light nuclei and decreases for very heavy nuclei due to Coulomb repulsion.

**Question Example 5.1**

Calculate the binding energy per nucleon for helium-4 ($^4$He) and $^{12}$C.

**Solution.**

$$\begin{aligned} BE(^4\mathrm{He}) &= \left[2m_p + 2m_n - M(^4\mathrm{He})\right] c^2 \approx 28.3\,\mathrm{MeV}, \\ BE(^4\mathrm{He})/4 &\approx 7.1\,\mathrm{MeV/nucleon}, \\ BE(^{12}\mathrm{C}) &= \left[6m_p + 6m_n - M(^{12}\mathrm{C})\right] c^2 \approx 92.2\,\mathrm{MeV}, \\ BE(^{12}\mathrm{C})/12 &\approx 7.7\,\mathrm{MeV/nucleon}. \end{aligned}$$

**THEORY**

The $Q$-value of a nuclear reaction is the net energy released or absorbed during the reaction, calculated as the difference in total mass-energy between the reactants and products. For a reaction $A + B \to C + D$, the $Q$-value is given by:

$$Q = \left[m_A + m_B - (m_C + m_D)\right] c^2. \tag{5.3}$$

A positive $Q$-value indicates an exothermic reaction, while a negative $Q$-value indicates an endothermic reaction.

### 5.1.2 Why Fusion Powers Stars

**PHYSICS INTUITION**

Stars gain energy when fusing light nuclei toward higher binding energy per nucleon. This trend reverses near iron, where further fusion is endothermic.

## 5.2 Hydrogen and Helium Burning

### 5.2.1 Proton-Proton (p-p) Chain

**FOUNDATION**

Representative net reaction:

$$4\,^1\mathrm{H} \to {}^4\mathrm{He} + 2e^+ + 2\nu_e + \text{energy}. \tag{5.4}$$

**Question Example 5.2**

Calculate the energy released in the proton-proton chain reaction.

**Solution.** The net reaction for the proton-proton chain is:

$$4\,^1\mathrm{H} \to {}^4\mathrm{He} + 2e^+ + 2\nu_e + \text{energy}. \tag{5.5}$$

The energy released can be calculated using the mass difference between the reactants and products:

$$\begin{aligned}\Delta m &= 4m_p - m_{^4\mathrm{He}} - 2m_e \\ &\approx 4 \times 1.007276\,\mathrm{u} - 4.002603\,\mathrm{u} - 2 \times 0.0005486\,\mathrm{u} \\ &\approx 0.0287\,\mathrm{u}.\end{aligned}$$

Converting this mass difference to energy using $E = \Delta mc^2$ and knowing that $1\,\mathrm{u} \approx 931.5\,\mathrm{MeV}/c^2$, we get:

$$\begin{aligned}E &= 0.0287\,\mathrm{u} \times 931.5\,\mathrm{MeV}/\mathrm{u} \\ &\approx 26.7\,\mathrm{MeV}.\end{aligned}$$

Thus, the energy released in the proton-proton chain reaction is approximately 26.7 MeV and carried away by neutrinos.

### 5.2.2 Helium Burning (Triple-Alpha Process)

**Foundation**

Representative net reaction:

$$3\,{}^{4}\mathrm{He} \to {}^{12}\mathrm{C} + \gamma + \text{energy}. \tag{5.6}$$

**Question Example 5.3**

Calculate the energy released in the triple-alpha process.

**Solution.** The net reaction for the triple-alpha process is:

$$3\,{}^{4}\mathrm{He} \to {}^{12}\mathrm{C} + \gamma + \text{energy}. \tag{5.7}$$

The energy released can be calculated using the mass difference between the reactants and products:

$$\begin{aligned} \Delta m &= 3m_{^4\mathrm{He}} - m_{^{12}\mathrm{C}} \\ &\approx 3 \times 4.002603\,\mathrm{u} - 12.000000\,\mathrm{u} \\ &\approx 0.007809\,\mathrm{u}. \end{aligned}$$

Converting this mass difference to energy using $E = \Delta mc^2$ and knowing that $1\,\mathrm{u} \approx 931.5\,\mathrm{MeV}/c^2$, we get:

$$\begin{aligned} E &= 0.007809\,\mathrm{u} \times 931.5\,\mathrm{MeV}/\mathrm{u} \\ &\approx 7.27\,\mathrm{MeV}. \end{aligned}$$

Thus, the energy released in the triple-alpha process is approximately 7.27 MeV.

## 5.3 CNO Cycle

The CNO cycle is a catalytic cycle that uses carbon, nitrogen, and oxygen as catalysts to convert hydrogen into helium. It dominates in stars more massive than the Sun due to its stronger temperature dependence compared to the p-p chain.

**Foundation**

Representative net reaction:

$$
\begin{aligned}
{}^{1}2\mathrm{C} + p &\rightarrow {}^{13}\mathrm{N} + \gamma, && (5.8)\\
{}^{13}\mathrm{N} &\rightarrow {}^{13}\mathrm{C} + e^{+} + \nu_e, && (5.9)\\
{}^{13}\mathrm{C} + p &\rightarrow {}^{14}\mathrm{N} + \gamma, && (5.10)\\
{}^{14}\mathrm{N} + p &\rightarrow {}^{15}\mathrm{O} + \gamma, && (5.11)\\
{}^{15}\mathrm{O} &\rightarrow {}^{15}\mathrm{N} + e^{+} + \nu_e, && (5.12)\\
{}^{15}\mathrm{N} + p &\rightarrow {}^{12}\mathrm{C} + {}^{4}\mathrm{He}. && (5.13)
\end{aligned}
$$

## 5.4 Advanced Burning Stages

**Foundation**

In massive stars, after the exhaustion of hydrogen and helium, heavier elements undergo fusion in successive burning stages: carbon burning, neon burning, oxygen burning, and silicon burning. Each stage produces heavier elements up to iron, which has the highest binding energy per nucleon.

**Theory**

The final stages of stellar evolution involve the synthesis of elements heavier than iron through neutron capture processes:

- **s-process** (slow neutron capture): Occurs in asymptotic giant branch (AGB) stars, where neutrons are captured at a rate slower than the beta decay of unstable nuclei.
- **r-process** (rapid neutron capture): Occurs in explosive environments such as supernovae and neutron star mergers, where neutrons are captured at a rate much faster than the beta decay of unstable nuclei.
- **p-process** (proton capture): Occurs in supernovae, where proton-rich isotopes are produced through proton capture and photodisintegration reactions.

## 5.5 Thermonuclear Reaction Rates

### 5.5.1 Coulomb Barrier and Tunneling

**FOUNDATION**

The **Gamov energy** is the minimum kinetic energy that two nuclei must have in order to overcome the Coulomb barrier and undergo fusion. It is given by the formula:

$$E_G = (\pi \alpha Z_1 Z_2)^2 2 m_r c^2 \tag{5.14}$$

where $m_r$ is the reduced mass of the two nuclei, $\alpha$ is the fine-structure constant, and $Z_1$ and $Z_2$ are the atomic numbers of the two nuclei.

**Question Example 5.4: The fusion of two protons**

Calculate the Gamov energy $E_G$ for the fusion of two protons.

**Solution.** For two protons, we have $Z_1 = Z_2 = 1$ and the reduced mass $m_r$ is approximately equal to the mass of a proton, $m_p$. The fine-structure constant $\alpha$ is approximately $1/137$. Plugging these values into the formula for the Gamov energy, we get:

$$\begin{aligned} E_G &= (\pi \alpha Z_1 Z_2)^2 2 m_r c^2 \\ &= (\pi \times \frac{1}{137} \times 1 \times 1)^2 \times 2 m_p c^2 \\ &= \left(\frac{\pi}{137}\right)^2 \times 2 m_p c^2 \\ &\approx 493\,\text{keV}. \end{aligned}$$

Thus, the Gamov energy for the fusion of two protons is approximately 493 keV.

**Question Example 5.5: The fusion of carbon and oxygen**

Calculate the Gamov energy $E_G$ for the fusion of carbon-12 ($^{12}$C) and oxygen-16 ($^{16}$O).

**Solution.** For carbon-12, we have $Z_1 = 6$ and for oxygen-16, we have $Z_2 = 8$. The reduced mass $m_r$ can be calculated using the formula:

$$m_r = \frac{m_1 m_2}{m_1 + m_2}, \tag{5.15}$$

where $m_1$ and $m_2$ are the masses of the carbon and oxygen nuclei, respectively. The mass of carbon-12 is approximately $12 m_p$ and the mass of oxygen-16 is approximately $16m_p$. Thus, we have:

$$\begin{aligned} m_r &= \frac{(12m_p)(16m_p)}{12m_p + 16m_p} \\ &= \frac{192m_p^2}{28m_p} \\ &= \frac{192}{28}m_p \\ &\approx 6.86m_p. \end{aligned}$$

Now we can calculate the Gamov energy using the formula:

$$\begin{aligned} E_G &= (\pi\alpha Z_1 Z_2)^2 2m_r c^2 \\ &= (\pi \times \frac{1}{137} \times 6 \times 8)^2 \times 2 \times 6.86m_p c^2 \\ &= \left(\frac{48\pi}{137}\right)^2 \times 2 \times 6.86m_p c^2 \\ &\approx 8.7\,\text{MeV}. \end{aligned}$$

Thus, the Gamov energy for the fusion of carbon-12 and oxygen-16 is approximately 8.7 MeV.

**FOUNDATION**

The electrostatic interaction (Coulomb barrier) between two positively charged nuclei is given by:

$$V_C(r) = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r}. \tag{5.16}$$

Classically, nuclei with $E < V_C$ cannot fuse as they do not have enough kinetic energy to overcome the Coulomb barrier.
Quantum tunneling gives non-zero penetration probability $P$ for $E < V_C$. The probability of tunneling through the Coulomb barrier can be approximated by the formula:

$$P \propto \exp\left[-\sqrt{\frac{E_G}{E}}\right], \tag{5.17}$$

where $E_G$ is the Gamow energy.

## 5.6 Thermonuclear Reaction Rates

**THEORY**

The cross-section $\sigma v$ for a nuclear reaction can be expressed as:

$$\langle \sigma v \rangle = \left( \frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp \left[ -\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}} \right] dE. \tag{5.18}$$

The integrand peaks at the Gamow energy window, balancing high-energy tail abundance and tunneling probability.

**Question Example 5.6**

Find the peak of the integrand in the expression for $\langle \sigma v \rangle$ to determine the most effective energy for nuclear reactions at a given temperature.

**Solution.** To find the peak of the integrand, we need to maximize the function:

$$f(E) = \exp \left[ -\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}} \right]. \tag{5.19}$$

Taking the natural logarithm of $f(E)$, we get:

$$\ln f(E) = -\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}}. \tag{5.20}$$

To find the maximum, we take the derivative with respect to $E$ and set it to zero:

$$\frac{d}{dE} \ln f(E) = -\frac{1}{k_B T} + \frac{1}{2} \sqrt{\frac{E_G}{E^3}} = 0. \tag{5.21}$$

Rearranging this equation gives:

$$\frac{1}{k_B T} = \frac{1}{2} \sqrt{\frac{E_G}{E^3}}. \tag{5.22}$$

Squaring both sides, we get:

$$\frac{1}{(k_B T)^2} = \frac{1}{4} \frac{E_G}{E^3}. \tag{5.23}$$

Rearranging this equation gives:

$$E^3 = \frac{1}{4} E_G (k_B T)^2. \tag{5.24}$$

Taking the cube root of both sides, we find the most effective energy for nuclear reactions at a given temperature:

$$E_0 = \left( \frac{1}{4} E_G (k_B T)^2 \right)^{1/3} = \left( \frac{E_G}{4} \right)^{1/3} (k_B T)^{2/3}. \tag{5.25}$$

Thus, the most effective energy for nuclear reactions at a given temperature is approximately $E_0 = \left( \frac{E_G}{4} \right)^{1/3} (k_B T)^{2/3}$.

**THEORY**

$$R_{AB} \propto T^a \tag{5.26}$$

where $a = \sqrt[3]{\frac{E_G}{4kT}}$ is a positive exponent that depends on the specific reaction and the temperature range.

**Question Example 5.7**

Calculate the temperature dependence of the proton-proton (p-p) chain reaction rate.

**Solution.** The proton-proton chain reaction has a Gamow energy of approximately $E_G \approx 493\,\text{keV}$. The temperature dependence can be approximated by:

$$R_{pp} \propto T^{\sqrt[3]{\frac{E_G}{4kT}}}. \tag{5.27}$$

At typical stellar core temperatures of around $T \approx 1.5 \times 10^7\,\text{K}$, the exponent $a$ can be calculated as:

$$a = \sqrt[3]{\frac{493\,\text{keV}}{4k(1.5 \times 10^7\,\text{K})}} \approx 4.5. \tag{5.28}$$

Thus, the reaction rate for the proton-proton chain scales approximately as $R_{pp} \propto T^{4.5}$ at typical stellar core temperatures.

CHAPTER 6
# Stellar Models

## 6.1 Stellar Structure Models

Main assumptions are made in simple stellar models:

- The star is in hydrostatic equilibrium.
- The star is in thermal equilibrium.
- The star is spherically symmetric.
- The star is non-rotating and has no magnetic fields.
- The star is composed of an ideal gas.

**THEORY**

These assumptions allow us to derive the basic equations governing the structure of stars, such as the equations of hydrostatic equilibrium, mass conservation, energy generation, and energy transport. While these models are simplified, they provide valuable insights into the fundamental properties and behaviors of stars.

$$\text{Mass conservation}: \quad \frac{dm(r)}{dr} = 4\pi r^2 \rho(r) \tag{6.1}$$

$$\text{Hydrostatic equilibrium}: \quad \frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2} \tag{6.2}$$

$$\text{Thermal equilibrium}: \quad L(r) = \int_0^r 4\pi r'^2 \rho(r') \epsilon(r') dr' \tag{6.3}$$

$$\text{Energy transport}: \quad \frac{dT}{dr} = -\frac{3\kappa(r)\rho(r)L(r)}{16\pi a c T^3 r^2} \tag{6.4}$$

**Question Example 6.1: Estimating Central Pressure of a Star**

onsider a star with a mass of $M = 1.5M_\odot$ and a radius of $R = 2R_\odot$. Assuming the star is in hydrostatic equilibrium and composed of an ideal gas, we can estimate the central pressure $P_c$ using the equation of hydrostatic equilibrium:

$$P_c \approx \frac{GM^2}{8\pi R^4} \tag{6.5}$$

Substituting the values for the Sun, we get:

$$P_c \approx \frac{G(1.5M_\odot)^2}{8\pi(2R_\odot)^4} \approx 1.5 \times 10^{16}\,\text{Pa} \tag{6.6}$$

**THEORY**

By expressing the stellar structure equations in terms of mass $m$ instead of radius $r$, we can derive a set of equations that are often more convenient for numerical integration. The equations become:

$$\frac{dr}{dm} = \frac{1}{4\pi r^2 \rho} \tag{6.7}$$

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4} \tag{6.8}$$

$$\frac{dL}{dm} = \epsilon \tag{6.9}$$

$$\frac{dT}{dm} = -\frac{3\kappa L}{16\pi a c T^3 r^4} \tag{6.10}$$

### 6.1.1 Mass Range of Stars

The mass range of stars is typically defined as follows:

- **Low-mass stars**: $M < 0.5 M_\odot$ (e.g., red dwarfs)
- **Intermediate-mass stars**: $0.5 M_\odot < M < 8 M_\odot$ (e.g., Sun-like stars)
- **High-mass stars**: $M > 8 M_\odot$ (e.g., O and B type stars)
- **Brown dwarfs**: $M < 0.08 M_\odot$ (substellar objects that do not sustain hydrogen fusion)

### 6.1.2 Mass-Luminosity relationship

The mass-luminosity relationship is an empirical relation that describes how the luminosity of a star depends on its mass.

**FOUNDATION**

For main-sequence stars, the luminosity $L$ is approximately proportional to a power of the mass $M$:

$$L \propto M^\alpha \tag{6.11}$$

where $\alpha$ is a constant that varies depending on the mass range of the star. For stars with masses similar to the Sun, $\alpha$ is typically around 3.5. For very low-mass stars, $\alpha$ can be closer to 2, while for very high-mass stars, $\alpha$ can be greater than 3.5.

CHAPTER 7

# Chapter 7: Types of Stars

## 7.1 Hertzsprung-Russell (HR) Diagram

PHYSICS INTUITION

The Hertzsprung-Russell (HR) diagram is a fundamental tool in astrophysics that plots stars according to their **luminosity** (or absolute magnitude) against their **surface temperature** (or spectral type). The HR diagram reveals distinct groups of stars, such as the main sequence, giants, and white dwarfs, and provides insights into stellar evolution and properties.

### 7.1.1 Spectral Classification

Apart from direct use of $L$, we can also estimate $T_{\mathrm{eff}}$ from the star's spectral type, which is determined by its surface temperature and spectral features.

THEORY

Stars are classified into spectral types O, B, A, F, G, K, M based on their surface temperatures and spectral features. O-type stars are the hottest and most massive, while M-type stars are the coolest and least massive. Each spectral type corresponds to a specific range of surface temperatures, which can be used to estimate $T_{\mathrm{eff}}$ if the spectral type is known.

## 7.2 Main Sequence Stars

## 7.3 White Dwarfs

THEORY

White dwarfs are supported by **electron degeneracy pressure**, which arises from the Pauli exclusion principle. The maximum mass of a white dwarf is given by the Chandrasekhar limit $M_{\mathrm{Ch}}$, approximately $1.4 M_{\odot}$.

**Question Example 7.1**

Calculate the Chandrasekhar limit for a white dwarf composed of carbon and oxygen, assuming that the electrons are non-relativistic and the star is supported by electron degeneracy pressure.

**Solution.** The Chandrasekhar limit can be derived by balancing the gravitational force with the electron degeneracy pressure. The maximum mass $M_{\text{Ch}}$ can be expressed as:

$$M_{\text{Ch}} = \frac{5.83}{\mu_e^2} M_\odot, \tag{7.1}$$

where $\mu_e$ is the mean molecular weight per electron. For a white dwarf composed of carbon and oxygen, $\mu_e \approx 2$. Substituting this value, we get:

$$M_{\text{Ch}} = \frac{5.83}{2^2} M_\odot = \frac{5.83}{4} M_\odot \approx 1.46 M_\odot. \tag{7.2}$$

Thus, the Chandrasekhar limit for a white dwarf composed of carbon and oxygen is approximately $1.46 M_\odot$.

## 7.4 Supernovae

CHAPTER 8

# Stellar Evolution

## 8.1 Early Stages of Stellar Evolution

## 8.2 Late Stages of Stellar Evolution

### 8.2.1 Helium Flash

### 8.2.2 Giants and Supergiants

### 8.2.3 Type II Supernovae

## 8.3 Neutron Stars

### 8.3.1 Tolman-Oppenheimer-Volkoff (TOV) Equation

**THEORY**

The **Tolman-Oppenheimer-Volkoff (TOV) equation** describes the structure of a spherically symmetric body in hydrostatic equilibrium in general relativity. It is given by:

$$\frac{dP(r)}{dr} = -\frac{G\left(\rho(r) + \frac{P(r)}{c^2}\right)\left(m(r) + 4\pi r^3 \frac{P(r)}{c^2}\right)}{r^2\left(1 - \frac{2Gm(r)}{rc^2}\right)} \tag{8.1}$$

where $P(r)$ is the pressure, $\rho(r)$ is the energy density, $m(r)$ is the mass enclosed within radius $r$, $G$ is the gravitational constant, and $c$ is the speed of light. This equation is crucial for modeling neutron stars and other compact objects where relativistic effects are significant.

**Question Example 8.1: Newtonian Limit of the TOV Equation**

Show that in the Newtonian limit, the TOV equation reduces to the Lane-Emden equation for a polytropic fluid.

**Solution.** In the Newtonian limit, we assume that the pressure is much less than the energy density ($P \ll \rho c^2$) and that the gravitational field is weak ($\frac{2Gm(r)}{rc^2} \ll 1$). Under these assumptions, the TOV equation simplifies to:

$$\frac{dP(r)}{dr} \approx -\frac{G\rho(r)m(r)}{r^2} \tag{8.2}$$

This is the hydrostatic equilibrium equation in Newtonian gravity. For a polytropic fluid, we can express the pressure as a function of density:

$$P = K\rho^{1+\frac{1}{n}} \tag{8.3}$$

where $K$ is a constant and $n$ is the polytropic index. Substituting this into the hydrostatic equilibrium equation and introducing dimensionless variables leads to the Lane-Emden equation, which describes the structure of polytropic stars in Newtonian gravity.

### 8.3.2 Mass-Radius Relation for Neutron Stars

**THEORY**

The mass-radius relation for neutron stars is determined by the equation of state (EOS) of neutron star matter. Generally, neutron stars have a mass-radius relation that can be approximated by:

$$R \propto M^{-1/3} \tag{8.4}$$

where $R$ is the radius of the neutron star and $M$ is its mass. This relation indicates that as the mass of a neutron star increases, its radius decreases, which is a consequence of the strong gravitational forces at play in these compact objects.

## 8.4 Chandrasekhar Limit of Neutron Stars

In this section, we will discuss the Chandrasekhar limit for neutron stars, which is the maximum mass that a neutron star can have before it collapses into a black hole. The Chandrasekhar limit for neutron stars is approximately $2.16M_\odot$, where $M_\odot$ is the mass of the Sun. This limit is determined by the balance between the degeneracy pressure of neutrons and the gravitational forces acting on the star. If a neutron star exceeds this limit, it can no longer support itself against gravitational collapse and will eventually form a black hole.

### 8.4.1 Pulsars/X-ray Brusters

PHYSICS INTUITION

**Pulsars** are rapidly rotating neutron stars that emit beams of electromagnetic radiation. As the neutron star rotates, these beams sweep across space, and if they point towards Earth, we observe regular pulses of radiation.
**X-ray bursters** are a type of binary system where a neutron star accretes matter from a companion star, leading to periodic bursts of X-rays due to thermonuclear explosions on the surface of the neutron star.

## 8.5 Black Holes

### 8.5.1 Schwarzschild Radius

THEORY

The **Schwarzschild radius** is the radius of the **event horizon** of a non-rotating black hole. It is given by:

$$r_s = \frac{2GM}{c^2} \tag{8.5}$$

where $G$ is the gravitational constant, $M$ is the mass of the black hole, and $c$ is the speed of light. Any object that is compressed within its Schwarzschild radius will become a black hole.

**Question Example 8.2: Schwarzschild Radius of a Collapsing Star**

Suppose we have a star with a mass of $10M_\odot$ (where $M_\odot$ is the mass of the Sun). Calculate the Schwarzschild radius of the black hole that would form if this star were to collapse.

**Solution.** First, we need to convert the mass of the star into kilograms. The mass of the Sun is approximately $1.989 \times 10^{30}$ kg, so:

$$M = 10M_\odot = 10 \times 1.989 \times 10^{30}\ \text{kg} = 1.989 \times 10^{31}\ \text{kg} \tag{8.6}$$

Using the formula for the Schwarzschild radius:

$$r_s = \frac{2GM}{c^2}, \tag{8.7}$$

we have

$$r_s = 29.49\ \text{km} \tag{8.8}$$

CHAPTER 9

# Practice Questions & Solutions

## 9.1 Chapter 1: Basic Concepts in Astrophysics

### 9.1.1 Questions

1.1 The star Gliese 710 is presently 62 lightyears from Earth, but Hipparcos data suggest that in about a million years it will pass within one lightyear of Earth. Its appearent magnitude is 9.7.

(a) What is the absolute magnitude of Gliese 710?

(b) What will be its apparent visual magnitude be in a million years as viewed from Earth?

1.2 Estimate the hydrodynamical free fall timescale for the Sun, a red giant of $M = M_\odot$ and $R = 100R_\odot$, and a white dwarf of $M = M_\odot$ and $R = 0.01R_\odot$ and $R = 100R_\odot$, and a white dwarf of $M = M_\odot$ and $R = 0.02R_\odot$.

1.3 Suppose that a star of mass $M$ and radius $R$ has a density distribution $\rho(r) = \rho_c(1 - \frac{r}{R})$, where $\rho_c$ is the density at the center of the star.

(a) Find an expression for $\rho_c$ in terms of $M$ and $R$.

(b) Calculate the mass $m(r)$ interior to radius $r$.

(c) Calculate the total gravitational binding energy of the star.

(d) Find an expression for the pressure $P(r)$ as a function of $r$, assuming that the star is in hydrostatic equilibrium and that the pressure at the surface ($r = R$) is zero.

(e) Assume that the material in the star is a **monatomic** ideal gas. Calculate the total internal energy of the star from $P(r)$, and show that the virial theorem is satisfied.

### 9.1.2 Solutions

**Solution 1.1: Absolute magnitude and apparent magnitude of Gliese** 710

(a) The absolute magnitude $M$ of a star can be calculated using the formula:

$$M = m - 5\log_{10}\left(\frac{d}{10}\right), \tag{9.1}$$

where $m$ is the apparent magnitude and $d$ is the distance to the star in parsecs. Since 1 lightyear is approximately 0.3064 parsecs, we can convert the distance to parsecs:

$$d = 62 \text{ lightyears} \times 0.3064 \text{ parsecs/lightyear} \approx 19.01 \text{ parsecs}. \tag{9.2}$$

Now we can calculate the absolute magnitude:

$$\begin{aligned} M &= 9.7 - 5\log_{10}\left(\frac{19.01}{10}\right) \\ &= 9.7 - 5\log_{10}(1.901) \\ &\approx 9.7 - 5 \times 0.279 = 9.7 - 1.395 = 8.305. \end{aligned}$$

Thus, the absolute magnitude of Gliese 710 is approximately 8.3.

(b) The apparent magnitude $m'$ of Gliese 710 in a million years can be calculated using the same formula, but with the new distance of 1 lightyear (or 0.3064 parsecs):

$$\begin{aligned} m' &= M + 5\log_{10}\left(\frac{d'}{10}\right) \\ &= 8.305 + 5\log_{10}(0.3064) \\ &\approx 8.305 + 5(-0.513) = 8.305 - 2.565 = 5.74. \end{aligned}$$

Thus, the apparent visual magnitude of Gliese 710 in a million years will be approximately 5.7.

**Solution 1.2: Hydrodynamical free fall timescale**

The hydrodynamical free fall timescale $t_{ff}$ can be estimated using the formula:

$$t_{ff} = \sqrt{\frac{3\pi}{32G\rho}}, \tag{9.3}$$

where $G$ is the gravitational constant and $\rho$ is the average density of the star. The average density can be calculated using the formula:

$$\rho = \frac{3M}{4\pi R^3}. \tag{9.4}$$

For the Sun, we have $M = M_\odot$ and $R = R_\odot$, so the average density is:

$$\rho_\odot = \frac{3M_\odot}{4\pi R_\odot^3}. \tag{9.5}$$

Substituting this into the formula for $t_{ff}$, we get:

$$t_{ff,\odot} = \sqrt{\frac{3\pi}{32G\rho_\odot}} = \sqrt{\frac{3\pi}{32G\frac{3M_\odot}{4\pi R_\odot^3}}} = \sqrt{\frac{4\pi^2 R_\odot^3}{32GM_\odot}} = \sqrt{\frac{\pi^2 R_\odot^3}{8GM_\odot}}. \tag{9.6}$$

For a red giant with $M = M_\odot$ and $R = 100R_\odot$, the average density is:

$$\rho_{RG} = \frac{3M_\odot}{4\pi(100R_\odot)^3} = \frac{3M_\odot}{4\pi 10^6 R_\odot^3} = \frac{3M_\odot}{4\pi 10^6 R_\odot^3}. \tag{9.7}$$

Substituting this into the formula for $t_{ff}$, we get:

$$t_{ff,RG} = \sqrt{\frac{3\pi}{32G\rho_{RG}}} = \sqrt{\frac{3\pi}{32G\frac{3M_\odot}{4\pi 10^6 R_\odot^3}}} = \sqrt{\frac{4\pi^2 10^6 R_\odot^3}{32GM_\odot}} = \sqrt{\frac{\pi^2 10^6 R_\odot^3}{8GM_\odot}}. \tag{9.8}$$

For a white dwarf with $M = M_\odot$ and $R = 0.02R_\odot$, the average density is:

$$\rho_{WD} = \frac{3M_\odot}{4\pi(0.02R_\odot)^3} = \frac{3M_\odot}{4\pi 8 \times 10^{-6} R_\odot^3} = \frac{3M_\odot}{4\pi 8 \times 10^{-6} R_\odot^3}. \tag{9.9}$$

Substituting this into the formula for $t_{ff}$, we get:

$$t_{ff,WD} = \sqrt{\frac{3\pi}{32G\rho_{WD}}} = \sqrt{\frac{3\pi}{32G\frac{3M_\odot}{4\pi 8\times 10^{-6} R_\odot^3}}} = \sqrt{\frac{4\pi^2 8 \times 10^{-6} R_\odot^3}{32GM_\odot}} = \sqrt{\frac{\pi^2 8 \times 10^{-6} R_\odot^3}{8GM_\odot}}. \tag{9.10}$$

Thus, the hydrodynamical free fall timescales for the Sun, the red giant, and the white dwarf can be calculated using the above formulas.

**Solution 1.3: Density distribution and hydrostatic equilibrium**

(a) To find an expression for $\rho_c$ in terms of $M$ and $R$, we can integrate the density distribution over the volume of the star to find the total mass:

$$M = \int_0^R 4\pi r^2 \rho(r) dr = \int_0^R 4\pi r^2 \rho_c \left(1 - \frac{r}{R}\right) dr. \tag{9.11}$$

Evaluating this integral, we get:

$$\begin{aligned} M &= 4\pi\rho_c \int_0^R r^2 \left(1 - \frac{r}{R}\right) dr \\ &= 4\pi\rho_c \left[\int_0^R r^2 dr - \frac{1}{R}\int_0^R r^3 dr\right] \\ &= 4\pi\rho_c \left[\frac{R^3}{3} - \frac{1}{R}\cdot\frac{R^4}{4}\right] = 4\pi\rho_c R^3 \left(\frac{1}{3} - \frac{1}{4}\right) = 4\pi\rho_c R^3 \cdot \frac{1}{12} = \frac{\pi}{3}\rho_c R^3. \end{aligned}$$

Solving for $\rho_c$, we get:

$$\rho_c = \frac{3M}{\pi R^3}. \tag{9.12}$$

(b) To calculate the mass $m(r)$ interior to radius $r$, we can integrate the density distribution from 0 to $r$:

$$m(r) = \int_0^r 4\pi r'^2 \rho(r') dr' = \int_0^r 4\pi r'^2 \rho_c \left(1 - \frac{r'}{R}\right) dr'. \tag{9.13}$$

Evaluating this integral, we get:

$$\begin{aligned} m(r) &= 4\pi\rho_c \int_0^r r'^2 \left(1 - \frac{r'}{R}\right) dr' \\ &= 4\pi\rho_c \left[\int_0^r r'^2 dr' - \frac{1}{R}\int_0^r r'^3 dr'\right] \\ &= 4\pi\rho_c \left[\frac{r^3}{3} - \frac{1}{R}\cdot\frac{r^4}{4}\right] = 4\pi\rho_c r^3 \left(\frac{1}{3} - \frac{r}{4R}\right). \end{aligned}$$

Substituting $\rho_c$ from part (a), we get:

$$m(r) = 4\pi \cdot \frac{3M}{\pi R^3} r^3 \left(\frac{1}{3} - \frac{r}{4R}\right) = \frac{12M}{R^3} r^3 \left(\frac{1}{3} - \frac{r}{4R}\right). \tag{9.14}$$

(c) To calculate the total gravitational binding energy of the star, we can use the formula:

$$U = -\int_0^R \frac{Gm(r)}{r} 4\pi r^2 \rho(r) dr. \tag{9.15}$$

Substituting $m(r)$ and $\rho(r)$, we get:

$$\begin{aligned} U &= -\int_0^R \frac{G}{r} \cdot \frac{12M}{R^3} r^3 \left(\frac{1}{3} - \frac{r}{4R}\right) \cdot \rho_c \left(1 - \frac{r}{R}\right) dr \\ &= -\int_0^R \frac{G}{r} \cdot \frac{12M}{R^3} r^3 \left(\frac{1}{3} - \frac{r}{4R}\right) \cdot \frac{3M}{\pi R^3} \left(1 - \frac{r}{R}\right) dr \\ &= -\frac{36GM^2}{\pi R^6} \int_0^R r^2 \left(\frac{1}{3} - \frac{r}{4R}\right) \left(1 - \frac{r}{R}\right) dr. \end{aligned}$$

Evaluating this integral will give the total gravitational binding energy of the star.

(d) To find an expression for the pressure $P(r)$ as a function of $r$, we can use the equation of hydrostatic equilibrium:

$$\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2}. \tag{9.16}$$

Substituting $m(r)$ and $\rho(r)$, we get:

$$\begin{aligned} \frac{dP}{dr} &= -\frac{G}{r^2} \cdot \frac{12M}{R^3} r^3 \left(\frac{1}{3} - \frac{r}{4R}\right) \cdot \rho_c \left(1 - \frac{r}{R}\right) \\ &= -\frac{12GM}{R^3} r \left(\frac{1}{3} - \frac{r}{4R}\right) \cdot \frac{3M}{\pi R^3} \left(1 - \frac{r}{R}\right) \\ &= -\frac{36GM^2}{\pi R^6} r \left(\frac{1}{3} - \frac{r}{4R}\right) \left(1 - \frac{r}{R}\right). \end{aligned}$$

Integrating this equation will give the pressure as a function of $r$.

(e) Assuming that the material in the star is a monatomic ideal gas, the internal energy $U$ can be calculated using the formula:

$$U = \frac{3}{2} \int_0^R 4\pi r^2 P(r) dr. \tag{9.17}$$

Substituting $P(r)$ from part (d), we can evaluate this integral to find the total internal energy of the star. To show that the virial theorem is satisfied, we need to verify that $2U + U_{grav} = 0$, where $U_{grav}$ is the gravitational binding energy calculated in part (c).

## 9.2 Chapter 2: Star Formation

### 9.2.1 Questions

## 9.3 Chapter 3: Thermodynamic Properties of Stars

### 9.3.1 Questions

1. Show that for the Maxwellian velocity distribution of particles in a gas leads to the ideal gas law $P = nkT$, where $P$ is the pressure, $n$ is the number density of particles, $k$ is Boltzmann's constant, and $T$ is the temperature.

### 9.3.2 Solutions

**Solution 3.1: Derivation of the ideal gas law from the Maxwellian velocity distribution**

The Maxwellian velocity distribution describes the distribution of speeds of particles in a gas. The number density of particles with velocity between $v$ and $v + dv$ is given by:

$$f(v)dv = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}} dv, \tag{9.18}$$

where $m$ is the mass of a particle, $k$ is Boltzmann's constant, and $T$ is the temperature. The pressure $P$ of the gas can be calculated using the kinetic theory of gases, which states that the pressure is related to the average kinetic energy of the particles:

$$P = \frac{1}{3} nm\langle v^2 \rangle, \tag{9.19}$$

where $n$ is the number density of particles and $\langle v^2 \rangle$ is the average of the square of the velocity. We can calculate $\langle v^2 \rangle$ using the Maxwellian velocity distribution:

$$\begin{aligned}\langle v^2 \rangle &= \int_0^\infty v^2 f(v) dv \\ &= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty v^4 e^{-\frac{mv^2}{2kT}} dv.\end{aligned}$$

To evaluate this integral, we can use the substitution $x = \frac{mv^2}{2kT}$, which gives $v^2 = \frac{2kT}{m}x$ and $dv = \sqrt{\frac{2kT}{m}} \frac{dx}{\sqrt{x}}$. Substituting these into the integral, we get:

$$\begin{aligned}\langle v^2 \rangle &= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty \left(\frac{2kT}{m}x\right)^2 e^{-x} \sqrt{\frac{2kT}{m}} \frac{dx}{\sqrt{x}} \\ &= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(\frac{2kT}{m}\right)^{5/2} \int_0^\infty x^{3/2} e^{-x} dx.\end{aligned}$$

The integral $\int_0^\infty x^{3/2} e^{-x} dx$ can be evaluated using the gamma function, which gives $\Gamma(5/2) = \frac{3\sqrt{\pi}}{4}$. Substituting this result back into the expression for $\langle v^2 \rangle$, we get:

$$\begin{aligned}\langle v^2 \rangle &= 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} \left(\frac{2kT}{m}\right)^{5/2} \cdot \frac{3\sqrt{\pi}}{4} \\ &= 3\frac{kT}{m}.\end{aligned}$$

Substituting this result back into the expression for pressure, we get:

$$P = \frac{1}{3} nm \langle v^2 \rangle = \frac{1}{3} nm \cdot 3\frac{kT}{m} = nkT. \quad (9.20)$$

Thus, we have shown that the Maxwellian velocity distribution leads to the ideal gas law $P = nkT$.

## 9.4 Chapter 4: Stellar Structure

### 9.4.1 Questions

1.

### 9.4.2 Solutions

## 9.5 Chapter 5: Nuclear Synthesis in Stars

### 9.5.1 Questions

5.1 Estimate the binding energy per nucleon of $^{56}$Fe and $^{4}$He, and explain why $^{56}$Fe is more stable than $^{4}$He.

5.2 Estimate the energy released per kilogram of matter by the sequence of reactions which fuses hydrogen into iron.

5.3 In the simplest model of $\alpha$ decay, the $\alpha$ particle is pre-formed and trapped within the nucleus by a potential similar to a Coulomb barrier. The mean decay rate $\lambda$ is the frequency $\nu$ (with which the $\alpha$ particle hits the barrier) times the probability of penetration of the barrier (quantum tunneling probability).

(a) Write the approximate expression for the decay rate in terms of $\nu$, $E_G$, and $E$ (where $E$ is the energy released by $\alpha$ decay).

(b) Find the Gamow energy of the $\alpha$ decay of $^{235}$U.

(c) Find the Gamow energy of the $\alpha$ decay of $^{239}$Pu.

(d) Assuming that the frequency $\nu$ is approximately the same for both nuclei, find the ratio

$$\frac{\lambda(^{239}\mathrm{Pu})}{\lambda(^{235}\mathrm{U})}$$

using $E(^{235}\mathrm{U}) = 4.68\mathrm{MeV}$ and $E(^{239}\mathrm{Pu}) = 5.24\mathrm{MeV}$. The lifetime of a radioactive species is inversely proportional to its decay rate. Given that the observed lifetime of $^{235}$U is $\tau(^{235}\mathrm{U}) = 7.1 \times 10^8 y$, estimate the lifetime of $^{239}$Pu. Compare your result with the experimental value

$$\tau(^{239}\mathrm{Pu}) = 2.41 \times 10^4 y$$

and comment on the accuracy of this simple model.

5.4 The **triple alpha process** is a set of nuclear reactions that occur in stars to produce carbon and other elements. Which of the following correctly describes the three steps of the triple alpha process, along with whether energy is absorbed or released during each step?

(a) $^4\mathrm{He} +^4\mathrm{He} \rightarrow^8\mathrm{Be}$

(b) $^8\mathrm{Be} +^4\mathrm{He} \rightarrow^{12}\mathrm{C}^*$

(c) $^{12}\mathrm{C}^* \rightarrow^{12}\mathrm{C} + \gamma$

5.5 In CNO cycle, the slowest reaction in the reaction chain is the proton capture by $^{14}$N,

$$^{14}\mathrm{N} + p \rightarrow^{15}\mathrm{O} + \gamma.$$

Estimate the temperature dependence $a$ of the CN cycle in the sun (at $T = 1.5 \times 10^7$ K).

### 9.5.2 Solutions

**Solution 5.1: Binding energy per nucleon**

The binding energy per nucleon can be calculated using the formula:

$$\text{Binding energy per nucleon} = \frac{Zm_p + Nm_n - M}{A}c^2, \tag{9.21}$$

where $Z$ is the number of protons, $N$ is the number of neutrons, $M$ is the mass of the nucleus, $A$ is the total number of nucleons (protons + neutrons), and $c$ is the speed of light. For $^4$He, we have $Z = 2$, $N = 2$, and $A = 4$. The mass of the helium nucleus is approximately $M = 4.002603u$. Substituting these values, we get:

$$\begin{aligned}\text{Binding energy per nucleon for } ^4\text{He} &\approx \frac{2 \times 1.007276u + 2 \times 1.008665u - 4.002603u}{4}c^2 \\ &\approx \frac{4.031882u - 4.002603u}{4}c^2 \\ &\approx \frac{0.029279u}{4}c^2 \\ &\approx 7.07 \text{ MeV}.\end{aligned}$$

For $^{56}$Fe, we have $Z = 26$, $N = 30$, and $A = 56$. The mass of the iron nucleus is approximately $M = 55.934936u$. Substituting these values, we get:

$$\begin{aligned}\text{Binding energy per nucleon for } ^{56}\text{Fe} &\approx \frac{26 \times 1.007276u + 30 \times 1.008665u - 55.934936u}{56}c^2 \\ &\approx \frac{26.189176u + 30.25995u - 55.934936u}{56}c^2 \\ &\approx \frac{56.449126u - 55.934936u}{56}c^2 \\ &\approx \frac{0.51419u}{56}c^2 \\ &\approx 9.18 \text{ MeV}.\end{aligned}$$

Thus, the binding energy per nucleon for $^{56}$Fe is approximately 9.18 MeV, while for $^4$He it is approximately 7.07 MeV. This means that $^{56}$Fe is more stable than $^4$He because it has a higher binding energy per nucleon, indicating that more energy would be required to break it apart into individual nucleons.

**Solution 5.2: Energy released by fusion of hydrogen into iron**

The energy released per kilogram of matter by the sequence of reactions which fuses hydrogen into iron can be estimated by calculating the difference in binding energy per nucleon between hydrogen and iron, and then multiplying by the number of nucleons per kilogram of matter. The binding energy per nucleon for hydrogen (proton) is approximately 0 MeV, while for iron it is approximately 9.18 MeV. The energy released per nucleon during the fusion process can be calculated as:

$$\text{Energy released per nucleon} = \text{Binding energy per nucleon of iron} - \text{Binding energy per nucleon}$$

To convert this energy to joules, we can use the conversion factor 1 MeV $= 1.60218 \times 10^{-13}$ J:

Energy released per nucleon $\approx 9.18$ MeV $\times 1.60218 \times 10^{-13}$ J/MeV $\approx 1.47 \times 10^{-12}$ J.

The number of nucleons per kilogram of matter can be calculated using the average mass of a nucleon, which is approximately $1.67 \times 10^{-27}$ kg:

$$\begin{aligned}\text{Number of nucleons per kg} &= \frac{1\text{ kg}}{1.67 \times 10^{-27}\text{ kg/nucleon}} \\ &\approx 5.99 \times 10^{26}\text{ nucleons/kg.}\end{aligned}$$

Finally, the energy released per kilogram of matter can be calculated as:

$$\begin{aligned}\text{Energy released per kg} &\approx 1.47 \times 10^{-12}\text{ J/nucleon} \times 5.99 \times 10^{26}\text{ nucleons/kg} \\ &\approx 8.80 \times 10^{14}\text{ J/kg.}\end{aligned}$$

Thus, the energy released per kilogram of matter by the sequence of reactions which fuses hydrogen into iron is approximately $8.80 \times 10^{14}$ joules.

**Solution 5.3: $\alpha$ decay and Gamow energy**

(a) The decay rate $\lambda$ can be expressed as:

$$\lambda = \nu e^{-2\pi\sqrt{\frac{E_G}{E}}}, \tag{9.22}$$

where $\nu$ is the frequency of collisions with the barrier, $E_G$ is the Gamow energy, and $E$ is the energy released by $\alpha$ decay.

(b) The Gamow energy $E_G$ for the $\alpha$ decay of $^{235}$U can be calculated using the formula:

$$E_G = (\pi\alpha Z_1 Z_2)^2 2m_r c^2, \tag{9.23}$$

where $Z_1 = 92$ (for uranium), $Z_2 = 2$ (for the alpha particle), and $m_r$ is the reduced mass of the system. The reduced mass can be approximated as:

$$m_r \approx \frac{m_{^{235}\mathrm{U}} m_\alpha}{m_{^{235}\mathrm{U}} + m_\alpha} \approx m_\alpha, \tag{9.24}$$

since the mass of uranium is much larger than that of the alpha particle. Substituting these values, we get:

$$\begin{aligned}E_G &\approx (\pi \times \frac{1}{137} \times 92 \times 2)^2 2m_\alpha c^2 \\ &\approx 25.7\,\text{MeV.}\end{aligned}$$

(c) Similarly, for the $\alpha$ decay of $^{239}$Pu, we have $Z_1 = 94$ (for plutonium) and $Z_2 = 2$. The reduced mass can again be approximated as $m_\alpha$. Substituting these values, we get:

$$\begin{aligned}E_G &\approx (\pi \times \frac{1}{137} \times 94 \times 2)^2 2m_\alpha c^2 \\ &\approx 27.0\,\text{MeV.}\end{aligned}$$

(d) The ratio of the decay rates can be calculated as follows:

$$\frac{\lambda(^{239}\mathrm{Pu})}{\lambda(^{235}\mathrm{U})} = \frac{\nu e^{-2\pi\sqrt{\frac{E_G(^{239}\mathrm{Pu})}{E(^{239}\mathrm{Pu})}}}}{\nu e^{-2\pi\sqrt{\frac{E_G(^{235}\mathrm{U})}{E(^{235}\mathrm{U})}}}} = e^{-2\pi\left(\sqrt{\frac{E_G(^{239}\mathrm{Pu})}{E(^{239}\mathrm{Pu})}}-\sqrt{\frac{E_G(^{235}\mathrm{U})}{E(^{235}\mathrm{U})}}\right)}.$$

Substituting the values of $E_G$ and $E$, we get:

$$\frac{\lambda(^{239}\mathrm{Pu})}{\lambda(^{235}\mathrm{U})} \approx e^{-2\pi\left(\sqrt{\frac{27.0}{5.24}}-\sqrt{\frac{25.7}{4.68}}\right)} \approx e^{-2\pi(2.27-2.35)} \approx e^{0.50} \approx 1.65.$$

Therefore, the decay rate of $^{239}$Pu is approximately 1.65 times that of $^{235}$U. Given that the lifetime $\tau$ is inversely proportional to the decay rate $\lambda$, we can estimate the lifetime of $^{239}$Pu as follows:

$$\tau(^{239}\mathrm{Pu}) \approx \frac{\tau(^{235}\mathrm{U})}{1.65} \approx \frac{7.1 \times 10^8 y}{1.65} \approx 4.3 \times 10^8 y. \tag{9.25}$$

Comparing this result with the experimental value of $\tau(^{239}\mathrm{Pu}) = 2.41 \times 10^4 y$, we see that the simple model significantly overestimates the lifetime of $^{239}$Pu. This discrepancy suggests that the model does not capture all the complexities of $\alpha$ decay, such as nuclear structure effects and other factors that can influence the decay rate.

**Solution 5.4: Triple alpha process**

The first two steps involve the absorption of energy to overcome the Coulomb barrier between the nuclei, while the final step releases energy in the form of a gamma photon as the excited carbon nucleus transitions to its ground state. Thus, the correct description of the three steps of the triple alpha process, along with whether energy is absorbed or released during each step, is:

(a) $^4\mathrm{He} +^4 \mathrm{He} \rightarrow^8 \mathrm{Be}$ (**energy absorbed**)

(b) $^8\mathrm{Be} +^4 \mathrm{He} \rightarrow^{12} \mathrm{C}^*$ (**energy absorbed**)

(c) $^{12}\mathrm{C}^* \rightarrow^{12} \mathrm{C} + \gamma$ (**energy released**)

**Solution 5.5: Temperature dependence of the CN cycle**

The Gamov energy for the proton capture by $^{14}$N can be calculated using the formula:

$$E_G = (\pi\alpha Z_1 Z_2)^2 2m_r c^2, \tag{9.26}$$

where $Z_1 = 7$ (for nitrogen), $Z_2 = 1$ (for the proton), and $m_r$ is the reduced mass of the system. The reduced mass can be approximated as:

$$m_r \approx \frac{m_{^{14}\mathrm{N}} m_p}{m_{^{14}\mathrm{N}} + m_p} \approx m_p, \tag{9.27}$$

since the mass of nitrogen is much larger than that of the proton. Substituting these values, we get:

$$\begin{aligned} E_G &\approx (\pi \times \frac{1}{137} \times 7 \times 1)^2 2 m_p c^2 \\ &\approx 3.5\,\mathrm{MeV}. \end{aligned}$$

The temperature dependence $a$ of the CN cycle can be estimated using the formula:

$$a = \sqrt[3]{\frac{E_G}{4k_B T}}, \tag{9.28}$$

where $k_B$ is the Boltzmann constant and $T$ is the temperature. Substituting the values, we get:

$$\begin{aligned} a &\approx \sqrt[3]{\frac{3.5\,\mathrm{MeV}}{4 \times 8.617 \times 10^{-5}\mathrm{eV/K} \times 1.5 \times 10^7\mathrm{K}}} \\ &\approx \sqrt[3]{\frac{3.5 \times 10^6\mathrm{eV}}{5.17 \times 10^3\mathrm{eV}}} \\ &\approx \sqrt[3]{676} \approx 8.8. \end{aligned}$$

Thus, the temperature dependence of the CN cycle in the sun is approximately $a \approx 8.8$.

## 9.6 Chapter 6: Stellar Models

### 9.6.1 Questions

1. Show that $F(r) = P + \frac{Gm^2}{8\pi r^4}$ is a decreasing function of $r$. Hence, find the lower limit of the central pressure $P_c$ of a star in terms of its mass $M$ and radius $R$.

### 9.6.2 Solutions

**Solution 6.1: Lower limit of central pressure**

To show that $F(r) = P + \frac{Gm^2}{8\pi r^4}$ is a decreasing function of $r$, we need to compute the derivative of $F(r)$ with respect to $r$ and show that it is negative. First, we compute the derivative of $F(r)$:

$$\begin{aligned}\frac{dF}{dr} &= \frac{dP}{dr} + \frac{d}{dr}\left(\frac{Gm^2}{8\pi r^4}\right) \\ &= \frac{dP}{dr} + \frac{G}{8\pi} \cdot \frac{d}{dr}\left(\frac{m^2}{r^4}\right).\end{aligned}$$

Next, we compute the derivative of $\frac{m^2}{r^4}$:

$$\begin{aligned}\frac{d}{dr}\left(\frac{m^2}{r^4}\right) &= \frac{d}{dr}(m^2) \cdot \frac{1}{r^4} + m^2 \cdot \frac{d}{dr}\left(\frac{1}{r^4}\right) \\ &= 2m \cdot \frac{dm}{dr} \cdot \frac{1}{r^4} - m^2 \cdot \frac{4}{r^5} \\ &= \frac{2m}{r^4} \cdot 4\pi r^2 \rho - \frac{4m^2}{r^5} \\ &= \frac{8\pi m\rho}{r^2} - \frac{4m^2}{r^5}.\end{aligned}$$

Substituting this back into the expression for $\frac{dF}{dr}$, we get:

$$\begin{aligned}\frac{dF}{dr} &= \frac{dP}{dr} + \frac{G}{8\pi}\left(\frac{8\pi m\rho}{r^2} - \frac{4m^2}{r^5}\right) \\ &= \frac{dP}{dr} + G \cdot \frac{m\rho}{r^2} - \frac{Gm^2}{2\pi r^5}.\end{aligned}$$

Using the equation of hydrostatic equilibrium, we know that $\frac{dP}{dr} = -\frac{Gm\rho}{r^2}$. Substituting this into the expression for $\frac{dF}{dr}$, we get:

$$\begin{aligned}\frac{dF}{dr} &= -\frac{Gm\rho}{r^2} + 2G \cdot \frac{m\rho}{r^2} - G \cdot \frac{m^2}{\pi r^5} \\ &= G \cdot \frac{m\rho}{r^2} - G \cdot \frac{m^2}{\pi r^5}.\end{aligned}$$

Since $m$ and $\rho$ are positive, and $r$ is also positive, we can see that $\frac{dF}{dr}$ is negative for all $r > 0$. Therefore, $F(r)$ is a decreasing function of $r$. To find the lower limit of the central pressure $P_c$, we evaluate $F(r)$ at $r = R$:

$$F(R) = P(R) + \frac{GM^2}{8\pi R^4}. \tag{9.29}$$

Since $F(r)$ is decreasing, we have $F(R) \leq \lim_{r\to 0} F(r) =: P_c$. Therefore, we can write:

$$P_c \geq P(R) + \frac{GM^2}{8\pi R^4}. \tag{9.30}$$

Assuming that the pressure at the surface of the star is negligible (i.e., $P(R) \approx 0$), we can simplify this to:

$$P_c \geq \frac{GM^2}{8\pi R^4}. \tag{9.31}$$

This gives us the lower limit of the central pressure of a star in terms of its mass $M$ and radius $R$.

## 9.7 Chapter 7: Types of Stars

### 9.7.1 Questions

## 9.8 Chapter 8: Stellar Evolution

### 9.8.1 Questions

1. Let $M_{ch,0}$ be the Chandrasekhar mass for a white dwarf with $Y_e = 1$ (pure hydrogen). Calculate the ratio of $M_{ch}$ to $M_{ch,0}$ for a white dwarf with $Y_e = 0.5$.
2. Chandrasekhar limit for a neutron star is approximately $2.16 M_{\odot}$. Assuming that the neutron star is composed of pure neutrons, calculate the corresponding $Y_e$ and compare it to the Chandrasekhar mass for a white dwarf with that $Y_e$.
3. Suppose a star has a mass of $10 M_{\odot}$ and is currently in the main sequence phase. Estimate the time it will take for the star to evolve into a red giant, assuming that the main sequence lifetime is approximately $10^{10}$ years for a star of this mass.

### 9.8.2 Solutions

**Solution 1.1: Ratio of Chandrasekhar masses for different $Y_e$ values**

The Chandrasekhar mass $M_{ch}$ is inversely proportional to the square of the mean molecular weight per electron $\mu_e$, which is related to $Y_e$ by $\mu_e = 1/Y_e$. Therefore, we can express $M_{ch}$ as:

$$M_{ch} = \frac{5.83}{\mu_e^2} M_{\odot} = 5.83 Y_e^2 M_{\odot}. \tag{9.32}$$

For $Y_e = 1$, we have:

$$M_{ch,0} = 5.83 \times 1^2 M_{\odot} = 5.83 M_{\odot}. \tag{9.33}$$

For $Y_e = 0.5$, we have:

$$M_{ch} = 5.83 \times (0.5)^2 M_{\odot} = 5.83 \times 0.25 M_{\odot} = 1.4575 M_{\odot}. \tag{9.34}$$

Thus, the ratio of $M_{ch}$ to $M_{ch,0}$ is:

$$\frac{M_{ch}}{M_{ch,0}} = \frac{1.4575 M_{\odot}}{5.83 M_{\odot}} \approx 0.25. \tag{9.35}$$

Therefore, the Chandrasekhar mass for a white dwarf with $Y_e = 0.5$ is approximately 25% of the Chandrasekhar mass for a white dwarf with $Y_e = 1$.

**Solution 1.2: Chandrasekhar limit for a neutron star and corresponding $Y_e$**

For a neutron star composed of pure neutrons, the mean molecular weight per electron $Y_e$ is effectively zero, since there are no electrons present. However, we can still calculate the corresponding $Y_e$ by considering the composition of the neutron star. In a neutron star, the matter is primarily composed of neutrons, with a small fraction of protons and electrons to maintain charge neutrality. The mean molecular weight per electron can be approximated as:

$$Y_e \approx \frac{n_e}{n_b}, \tag{9.36}$$

where $n_e$ is the number density of electrons and $n_b$ is the number density of baryons (neutrons and protons). For a neutron star, we can assume that $n_e \approx n_p$ (number density of protons) and $n_b \approx n_n + n_p$ (number density of neutrons plus protons). Since the neutron star is primarily composed of neutrons, we can approximate $n_n \gg n_p$, leading to $n_b \approx n_n$. Therefore, we have:

$$Y_e \approx \frac{n_p}{n_n} \approx 0, \tag{9.37}$$

which confirms that the mean molecular weight per electron is effectively zero for a neutron star composed of pure neutrons.

**Solution 1.3: Main sequence lifetime**

The main sequence lifetime of a star can be estimated using the formula:

$$t_{MS} \approx 10^{10} \left(\frac{M}{M_\odot}\right)^{-2.5} \text{ years}, \tag{9.38}$$

where $M$ is the mass of the star. For a star with a mass of $10M_\odot$, we can calculate the main sequence lifetime as follows:

$$t_{MS} \approx 10^{10} \left(\frac{10M_\odot}{M_\odot}\right)^{-2.5} \text{ years} = 10^{10} \times 10^{-2.5} \text{ years} = 10^{7.5} \text{ years}. \tag{9.39}$$

Thus, the star will evolve into a red giant in approximately $10^{7.5}$ years.

CHAPTER A

# Declaration of AI Assistance

## A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

**Author Verification:** All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

CHAPTER B

# Mathematical Tools & Derivations

## B.1 Useful Identities

### Trigonometric

$$\sin^2\theta + \cos^2\theta = 1, \quad \tan\theta = \sin\theta/\cos\theta, \quad \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B.$$

### Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

### Common Integrals

$$\int x^n\,dx = \tfrac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int e^{ax}\,dx = \tfrac{1}{a}e^{ax} + C,$$

$$\int \frac{dx}{x} = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C.$$

**Gamma–Zeta Integrals**

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\,dx = \Gamma(s)\zeta(s), \qquad \Re(s) > 1,$$

$$\int_0^\infty \frac{x^{s-1}}{e^x + 1}\,dx = \left(1 - 2^{1-s}\right)\Gamma(s)\zeta(s), \qquad \Re(s) > 0.$$

$$\int_0^\infty \frac{x^2}{e^x - 1}\,dx = 2\zeta(3), \qquad \int_0^\infty \frac{x^2}{e^x + 1}\,dx = \tfrac{3}{2}\zeta(3),$$

$$\int_0^\infty \frac{x^3}{e^x - 1}\,dx = \frac{\pi^4}{15} = 6\zeta(4), \qquad \int_0^\infty \frac{x^3}{e^x + 1}\,dx = \frac{7\pi^4}{120} = \tfrac{7}{8}\,6\zeta(4).$$

### Taylor Expansions

$$f(x) = f(a) + f'(a)(x-a) + \tfrac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$\begin{aligned}
\sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \tfrac{x^3}{3!} + \tfrac{x^5}{5!} - \cdots \\
\cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots \\
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \tfrac{x^2}{2!} + \cdots \\
\sqrt{1+x} &= \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \cdots
\end{aligned}$$

### Common Approximations

**Small-argument limits** ($|x| \ll 1$):

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}, \qquad \tan x \approx x,$$
$$\sinh x \approx x, \qquad \cosh x \approx 1 + \frac{x^2}{2}, \qquad \tanh x \approx x.$$

**Large positive argument** ($x \gg 1$):

$$\sinh x \approx \frac{1}{2}e^x, \qquad \cosh x \approx \frac{1}{2}e^x, \qquad \tanh x \approx 1.$$

**Large negative argument** ($x \ll -1$):

$$\sinh x \approx -\frac{1}{2}e^{-x}, \qquad \cosh x \approx \frac{1}{2}e^{-x}, \qquad \tanh x \approx -1.$$

## B.2 Relativistic Limits

The energy-momentum relation is $E^2 = p^2c^2 + m^2c^4$. So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

CHAPTER C
# Constants & Conversion Factors

## C.1 Physical Constants

| Quantity | Symbol | Value (SI) |
|---|---|---|
| Speed of light | $c$ | $2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$ |
| Gravitational constant | $G$ | $6.674 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ |
| Hubble constant today | $H_0$ | $67.8 \pm 0.77\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ |
| Cosmological constant | $\Lambda$ | $1.11 \times 10^{-52}\,\mathrm{m^{-2}}$ |
| Cosmological constant | $\Lambda$ | $2.036 \times 10^{-35}\,\mathrm{s^{-2}}$ |
| Planck constant | $\hbar$ | $1.055 \times 10^{-34}\,\mathrm{J\,s}$ |
| Boltzmann constant | $k_B$ | $1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ |
| Fermi constant | $G_F$ | $1.166 \times 10^{-5}\,\mathrm{GeV^{-2}}$ |
| Fine-structure constant today | $\alpha$ | $1/137.036$ |
| Planck mass | $M_{\mathrm{Pl}}$ | $2.176 \times 10^{-8}\,\mathrm{kg}$ |

## C.2 Conversion Factors

### Astronomical Units

| Conversion | Value |
|---|---|
| 1 astronomical unit (AU) | $1.496 \times 10^{11}\,\mathrm{m}$ |
| 1 light-year (ly) | $9.461 \times 10^{15}\,\mathrm{m}$ |
| 1 parsec (pc) | $3.086 \times 10^{16}\,\mathrm{m}$ |
| 1 parsec (pc) | $3.262\,\mathrm{ly}$ |
| 1 year (yr) | $3.156 \times 10^{7}\,\mathrm{s}$ |
| 1 Earth mass ($M_\oplus$) | $5.972 \times 10^{24}\,\mathrm{kg}$ |
| 1 Solar mass ($M_\odot$) | $1.989 \times 10^{30}\,\mathrm{kg}$ |
| Milky Way' mass($M_{\mathrm{MW}}$) | $1.5 \times 10^{12} M_\odot$ |
| 1 Earth radius ($R_\oplus$) | $6.371 \times 10^{6}\,\mathrm{m}$ |
| 1 solar radius ($R_\odot$) | $6.957 \times 10^{8}\,\mathrm{m}$ |
| Milky Way' radius | $15\,\mathrm{kpc}$ |
| 1 solar luminosity ($L_\odot$) | $3.828 \times 10^{26}\,\mathrm{W}$ |
| Milky Way' luminosity ($L_{\mathrm{MW}}$) | $3.5 \times 10^{10} L_\odot$ |

### Energy Units

| Conversion | Value |
|---|---|
| 1 eV | $1.602 \times 10^{-19}$ J |
| 1 electron mass ($m_e$) | $0.511\,\mathrm{MeV}/c^2$ |
| 1 electron mass ($m_e$) | $9.109 \times 10^{-31}$ kg |
| 1 proton mass ($m_p$) | $938.3\,\mathrm{MeV}/c^2$ |
| 1 proton mass ($m_p$) | $1.673 \times 10^{-27}$ kg |
| 1 neutron mass ($m_n$) | $939.6\,\mathrm{MeV}/c^2$ |
| 1 neutron mass ($m_n$) | $1.675 \times 10^{-27}$ kg |

## C.3 SI Unit Prefixes

| Prefix | Name | Factor |
|---|---|---|
| $10^{-12}$ | pico (p) | $10^{-12}$ |
| $10^{-9}$ | nano (n) | $10^{-9}$ |
| $10^{-6}$ | micro ($\mu$) | $10^{-6}$ |
| $10^{-3}$ | milli (m) | $10^{-3}$ |
| $10^{3}$ | kilo (k) | $10^{3}$ |
| $10^{6}$ | mega (M) | $10^{6}$ |
| $10^{9}$ | giga (G) | $10^{9}$ |
| $10^{12}$ | tera (T) | $10^{12}$ |

## C.4 Nuclei Binding Energies & Atomic Masses

| Nucleus | Atomic Mass (u) | B(MeV) | B/A (MeV/u) |
|---|---|---|---|
| $e^-$ | 0.00054858 | – | – |
| $p$ | 1.0072765 | – | – |
| $n$ | 1.0086649 | – | – |
| $^{1}$H | 1.0078250 | 0.000000 | 0.000000 |
| $^{2}$H | 2.0141018 | 2.224515 | 1.112258 |
| $^{3}$H | 3.0160493 | 8.481773 | 2.827258 |
| $^{3}$He | 3.0160293 | 7.718036 | 2.572679 |
| $^{4}$He | 4.0026032 | 28.295803 | 7.073951 |
| $^{6}$Li | 6.0151223 | 31.994603 | 5.332434 |
| $^{7}$Li | 7.0160040 | 39.244654 | 5.606379 |
| $^{8}$Be | 8.0053051 | 56.499667 | 7.062458 |
| $^{9}$Be | 9.0121821 | 58.165096 | 6.462788 |
| $^{12}$C | 12.0000000 | 92.162051 | 7.680171 |
| $^{13}$C | 13.0033550 | 97.108223 | 7.469863 |
| $^{14}$N | 14.0030740 | 104.658962 | 7.475640 |
| $^{15}$N | 15.0001090 | 115.492214 | 7.699481 |
| $^{16}$O | 15.9949150 | 127.619412 | 7.976213 |
| $^{17}$O | 16.9991320 | 131.762631 | 7.750743 |
| $^{18}$O | 17.9991610 | 139.806972 | 7.767054 |
| $^{20}$Ne | 19.9924400 | 160.645559 | 8.032278 |
| $^{24}$Mg | 23.9850420 | 198.257479 | 8.260728 |
| $^{28}$Si | 27.9769270 | 236.537285 | 8.447760 |
| $^{32}$S | 31.9720710 | 271.781333 | 8.493167 |
| $^{56}$Fe | 55.9349420 | 492.255833 | 8.790283 |

Where **B** is the total binding energy of the nucleus, and **B/A** is the average binding energy per nucleon.

Note: $u = 1.661 \times 10^{-27}\,\text{kg} = 931.5\,\text{MeV}/c^2$.

# REFERENCES